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UNDERGRADUATE THESIS

Corporate Network Design and Security Policy Implementation

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Undergraduate thesis titled "CORPORATE NETWORK DESIGN AND SECURITY POLICY IMPLEMENTATION", presented by Gülnaz Karaca, Furkan İyem, Kutlu Çağan Akın and submitted by our institution, has been evaluated in accordance with the relevant provisions of the EU Undergraduate Education and Teaching Regulations and the EU Institute of Science Education and Teaching Directive. It has been found worthy of defense and was deemed successful unanimously/majority in the thesis defense examination held on 15.06.2026.

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ETHICAL STATEMENT AND AUTHORSHIP DECLARATION

This report was prepared as a student project report for the FKG Bank corporate network and cyber security architecture. The network design, EVE-NG topology implementation, configuration work, troubleshooting process, testing evidence and interpretation were prepared by Kutlu Çağan Akın, Furkan İyem and Gülnaz Karaca as a team project. External academic papers, standards, vendor concepts and online references used to support the technical background are cited in the References section.

The report follows the thesis-writing logic provided by the uploaded Ege University thesis writing guide: A4 layout, Times New Roman body text, structured sections, tables, figure captions and a references section. It is written in English while preserving device names and network labels used in the EVE-NG topology.

ÖZET

Bu proje, FKG Bank için güvenli bir kurumsal bankacılık ağının tasarlanması, uygulanması ve doğrulanmasını sunmaktadır. Ağ, EVE-NG ortamında gerçekleştirilmiş olup bir merkez ofis (Headquarters), bir ana perakende şubesi (Main Retail Branch), iki uydu şube (Satellite Branch) ve bir uzaktan çalışan (Remote Worker) segmentinden oluşmaktadır. Tasarımın amacı yalnızca cihazlar arasında erişilebilirlik sağlamak değil, aynı zamanda bağlantı, güvenlik segmentasyonu, şubeler arası şifreli iletişim, DMZ izolasyonu ve merkezi izleme mekanizmalarının birlikte değerlendirildiği gerçekçi bir bankacılık ortamı oluşturmaktır.

Nihai topoloji; Merkez Ofis (10.1.0.0/16), Ana Perakende Şube (10.2.0.0/16), Uydu Şube 1 (10.3.0.0/16), Uydu Şube 2 (10.4.0.0/16) ve Uzaktan Çalışan Ağı'ndan (10.1.220.0/24) oluşmaktadır. Tüm lokasyonlar merkezi bir İnternet/WAN bulutu üzerinden birbirine bağlanmıştır. Merkez ofiste çevre güvenlik geçidi olarak Cisco ASA kullanılmış, Ana Perakende Şube'de şube tipi ASA güvenlik duvarı konumlandırılmış, diğer şubelerde ise yönlendirici tabanlı bağlantı yapısı tercih edilmiştir. Tüm şubeler ile merkez arasında Site-to-Site IPsec VPN tünelleri kurulmuş, uzaktan çalışan için ise Remote Access VPN modeli uygulanmıştır. VPN geçiş altyapısı üzerinde dinamik yönlendirme protokolü olarak OSPF kullanılmış ve VLAN segmentasyonu sayesinde vezne (teller), ATM, CCTV, misafir ağı, ses (voice), yazıcı, yönetim, DMZ ve sunucu trafiği birbirinden ayrılmıştır.

Güvenlik mimarisi; Zero Trust, Least Privilege (En Az Yetki) ve Defense-in-Depth (Katmanlı Savunma) prensipleri temel alınarak oluşturulmuştur. Misafir ve ATM ağları bilinçli olarak kısıtlanmış, DMZ servisleri iç ağ kaynaklarından izole edilmiş, şubeler arası iletişim merkez ofis üzerinden yönlendirilmiş ve güvenlik duvarı ile ACL kuralları kullanılarak izin verilen trafik açık şekilde tanımlanmıştır. Projenin doğrulanması kapsamında arayüz kontrolleri, VLAN testleri, istemci–ağ geçidi erişim testleri, merkez–şube erişim testleri, şubeler arası iletişim testleri, IPsec Security Association (SA) kontrolleri, OSPF komşuluk doğrulamaları, ASA packet-tracer analizleri ve kısıtlı segmentler için negatif güvenlik testleri gerçekleştirilmiştir. Sorun giderme çalışmaları sırasında Cisco ASA üzerindeki NAT route-lookup davranışı, EVE-NG yönlendirici imajlarında gözlemlenen CEF yönlendirme problemleri ve DMZ ağındaki IP adresi çakışmaları çözülmüştür. Sonuç olarak ortaya çıkan mimari, yalnızca erişilebilirliği hedefleyen bir ağ topolojisinden ziyade, rol tabanlı erişim kontrolü ve güvenlik ilkeleriyle desteklenen, test edilebilir ve gerçekçi bir bankacılık ağı modeli sunmaktadır.

Anahtar Kelimeler: EVE-NG, bankacılık ağı, IPsec VPN, VLAN segmentasyonu, Cisco ASA, OSPF, DMZ, Zero Trust, Least Privilege, Syslog, ağ güvenliği.

ABSTRACT

This project presents the design, implementation and validation of a secure enterprise banking network named FKG Bank. The network was implemented in EVE-NG and structured around one headquarters site, one main retail branch, two satellite branches and one remote worker segment. The design objective was not limited to providing reachability between devices. Instead, the project aimed to model a banking environment where connectivity, security segmentation, encrypted inter-site communication, DMZ isolation and centralized monitoring are evaluated together.

The final topology contains Headquarters (10.1.0.0/16), Main Retail Branch (10.2.0.0/16), Satellite Branch 1 (10.3.0.0/16), Satellite Branch 2 (10.4.0.0/16) and a Remote Worker network (10.1.220.0/24). The sites are connected through a central Internet/WAN cloud. HQ uses Cisco ASA as the perimeter security gateway, while BR1 uses a branch ASA and BR2/BR3 use router-based branch connectivity. Site-to-site IPsec VPN tunnels connect all branches to HQ, while a remote access VPN model is used for the remote worker. OSPF is used as the internal dynamic routing protocol over the VPN transit path, and VLAN segmentation separates teller, ATM, CCTV, guest, voice, printer, management, DMZ and server-related traffic.

The security architecture is based on Zero Trust, Least Privilege and defense-in-depth principles. Guest and ATM networks are intentionally restricted; DMZ services are isolated from internal resources; branch-to-branch communication is routed through HQ; and firewall/ACL rules are used to define permitted traffic explicitly. The project was validated through interface checks, VLAN tests, endpoint-to-gateway pings, HQ-to-branch reachability tests, branch-to-branch tests, IPsec SA and OSPF neighbor checks, ASA packet-tracer analysis and negative tests for restricted segments. Troubleshooting activities included resolving NAT route-lookup behavior on Cisco ASA, CEF forwarding issues in the EVE-NG router images and duplicate IP conflicts in the DMZ. The resulting architecture demonstrates a testable, role-based banking network model rather than a simple reachability-only topology.

Keywords: EVE-NG, banking network, IPsec VPN, VLAN segmentation, Cisco ASA, OSPF, DMZ, Zero Trust, Least Privilege, Syslog, network security.

ACKNOWLEDGMENTS

The project team would like to express its sincere gratitude to our project advisor, Prof. Dr. Orhan Dağdeviren, for the valuable guidance, constructive feedback, and continuous support provided throughout the development of this project.

We are especially thankful for the encouragement to approach the project not only as a technical implementation exercise but also as an opportunity to understand and document the complete engineering process, including network design decisions, security considerations, troubleshooting activities, testing procedures, and the practical challenges encountered during implementation.

The knowledge and experience gained while designing, implementing, securing, and validating the FKG Bank network infrastructure have significantly contributed to our understanding of enterprise networking and cybersecurity principles. We also appreciate the support provided by the Department of Computer Engineering and the resources made available for conducting this project.

Finally, we would like to thank everyone who contributed directly or indirectly to the successful completion of this work.

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ABBREVIATIONS

Abbreviation	Meaning
ACL	Access Control List
ASA	Adaptive Security Appliance
CEF	Cisco Express Forwarding
CCTV	Closed-Circuit Television
DB	Database
DMZ	Demilitarized Zone
EVE-NG	Emulated Virtual Environment Next Generation
HQ	Headquarters
IKE	Internet Key Exchange
IPsec	Internet Protocol Security
NAT	Network Address Translation
OSPF	Open Shortest Path First
RA VPN	Remote Access VPN
SOC	Security Operations Center
SVI	Switch Virtual Interface
VLAN	Virtual Local Area Network
VLSM	Variable Length Subnet Masking
VPN	Virtual Private Network
WAN	Wide Area Network

1. INTRODUCTION

Enterprise banking networks must provide availability, confidentiality, integrity and controlled access at the same time. In a banking environment, connecting every endpoint to every other endpoint is not a valid success criterion. Teller workstations, ATM devices, CCTV cameras, guest wireless devices, DMZ servers and administrative computers have different trust levels and operational purposes. Therefore, a secure banking network must combine routing, switching, VPN encryption, firewall rules, VLAN segmentation and monitoring into a single coherent design.

The FKG Bank project was developed to model such an environment in EVE-NG. The final design consists of a central Headquarters site, one Main Retail Branch, two Satellite Branches and a Remote Worker connection model. The sites are linked over an Internet/WAN cloud using private WAN addresses in the 192.168.100.0/24 range. HQ acts as the central hub for encrypted branch connectivity. The design uses IPsec VPN tunnels, OSPF routing, Cisco ASA firewall policies and VLAN-based segmentation to emulate a secure enterprise banking infrastructure.

The project vision can be summarized as "not only connecting, but controlling". This statement is important because several tests were intentionally expected to fail. For instance, guest devices should not reach internal corporate subnets, ATM devices should not freely reach teller workstations or arbitrary HQ systems, and public-facing DMZ servers should not have unrestricted access to the inside network. The network is therefore evaluated through both positive tests, where authorized traffic must work, and negative tests, where unauthorized traffic must be blocked.

The final report revises the previous documentation according to the final topology photograph, the project presentation and the updated device inventory. Earlier draft references to a 10.5.0.0/16 BR3 block were removed. The final Satellite Branch 2 network is documented as 10.4.0.0/16, consistent with the final topology and the provided project list.

1.1 Project Aim

The aim of this project is to design, implement and validate a secure banking network architecture in EVE-NG. The implementation must demonstrate secure HQ-to-branch connectivity, role-based VLAN segmentation, DMZ isolation, remote worker access, controlled branch-to-branch communication and centralized logging readiness.

1.2 Scope

The scope includes logical network design, IP addressing, VLAN mapping, device-level configuration summaries, site-to-site VPN design, OSPF-based routing, DMZ and server zone separation, branch security policies, remote worker VPN concept, monitoring/SOC architecture and test validation. The report focuses on the final implemented topology rather than all theoretical VLANs listed in the scalable design catalog. VLANs appearing in the slide deck but not populated by active endpoints are treated as design-reserve VLANs.

1.3 Team Members

Kutlu Çağan AKIN

- Student Number: 05210000260
- Role in Project: HQ, Data Center, ASA

Furkan İYEM

- Student Number: 05220000316
- Role in Project: Main Retail, BR2, Branch Security

Gülnaz KARACA

- Student Number: 05210000266
- Role in Project: BR3, VPN, Syslog, Remote Worker, Documentation

2. LITERATURE AND TECHNICAL BACKGROUND

The design is grounded in established enterprise networking concepts: hierarchical network design, VLAN segmentation, perimeter firewalling, site-to-site VPN, remote access VPN, dynamic routing, centralized logging and defense-in-depth. The project does not claim to invent a new protocol. Its contribution is the integration of standard mechanisms into a coherent, role-based banking scenario and the validation of both permitted and denied traffic paths.

2.1 Enterprise Hierarchical Network Design

Hierarchical enterprise network design separates the network into logical layers such as core, distribution and access. This separation makes large topologies easier to understand, scale and troubleshoot. In the FKG Bank topology, HQ follows the hierarchical model more strongly than the smaller branches. HQ contains core switching, data center access, user access and DMZ access. Branches use smaller access-layer designs because their endpoint count is lower.

2.2 IPsec VPN for Inter-Site Security

IPsec provides network-layer protection for traffic crossing untrusted transport networks. The uploaded reference paper on virtual enterprise networks describes IPsec as a framework for private communication over public networks and identifies VPN as a common network-layer security control (Rosu et al., 2012). This project follows the same architectural idea: branch networks are not treated as directly trusted because they traverse a shared Internet/WAN cloud. Site-to-site IPsec tunnels therefore protect branch-to-HQ traffic.

2.3 VLAN Segmentation and Lateral Movement Control

VLANs are used to create separate broadcast domains over shared switching infrastructure. In this project, segmentation is not only an administrative convenience; it is part of the security model. Teller, ATM, CCTV, printer, voice, guest and management traffic are separated so that a compromise in one segment does not automatically allow movement into another segment. The project presentation explicitly defines VLAN segmentation as a defense-in-depth layer against lateral movement.

2.4 Zero Trust and Least Privilege

Zero Trust is interpreted operationally as the removal of implicit trust between network segments. Least Privilege means that users, devices and services receive only the connectivity required for their role. In the project presentation, this principle is expressed as the idea that the goal is not merely to connect devices, but to control and restrict traffic. The report preserves this principle in its security policy, test matrix and discussion.

2.5 Monitoring and Auditability

Banking networks require visibility. Firewall events, VPN connections, denied packets and authentication-related events should not disappear without trace. The project therefore includes an HQ-SYSLOG-SRV concept and treats centralized logging as a SOC-oriented layer. Although the lab environment does not implement a full SIEM platform, it prepares the topology for centralized logging and audit-friendly event collection.

2.6 Design Science Perspective

The work can be interpreted as a design science project: a practical artifact is created, configured, tested and evaluated against requirements. The artifact is the EVE-NG banking network topology and its related configuration and validation evidence. The evaluation relies on connectivity tests, route and tunnel state checks, ACL enforcement checks and troubleshooting records.

3. METHODOLOGY

The methodology followed a design-build-test-refine cycle. First, banking-specific requirements were identified. Second, the logical topology and IP address plan were created. Third, devices were placed and connected in EVE-NG. Fourth, VLANs, gateways, VPN tunnels, OSPF and security rules were configured. Fifth, the design was validated through step-by-step tests. Finally, detected issues were corrected and documented.

The testing process was intentionally layered. Local interface status and VLAN membership were verified before VPN and end-to-end tests. This avoided treating every ping failure as a VPN problem. For example, DMZ connectivity failed because the PC and Web Server had duplicate IP addresses; the correct solution was not to change VPN configuration, but to remove the IP conflict and clear ARP state. Similarly, HQ-to-BR2 endpoint traffic initially failed because of NAT route-lookup and CEF forwarding behavior, not because the branch endpoints were incorrectly addressed.

The final report also distinguishes between active implementation and scalable design intent. The slide deck contains a broader VLAN catalog for a production-like banking network. The actual EVE-NG topology implements the subset that is visible in the final topology photograph and necessary for the live demonstration. Reserved VLANs are still documented as design capacity but are not presented as fully populated endpoint zones unless they exist in the final device list.

3.1 Design Steps

1. Define HQ, BR1, BR2, BR3 and Remote Worker sites with unique IP blocks.
2. Assign WAN addresses to the Internet cloud-facing interfaces.

3. Create VLANs for endpoint roles such as Teller, ATM, CCTV, Guest, Voice, Printer, Management and DMZ.
4. Configure core/access switching and router-on-a-stick or ASA-based segmentation where applicable.
5. Build site-to-site IPsec tunnels from branches to HQ.
6. Run OSPF over tunnel links and internal branch networks.
7. Define ASA NAT exemption and ACL policies.
8. Validate positive and negative traffic flows.
9. Document troubleshooting outcomes and final lessons learned.

3.2 Validation Philosophy

Validation does not mean that every ping succeeds. In a secure banking design, some flows must be blocked. Therefore, validation includes positive tests for authorized reachability and negative tests for intentional restrictions. This approach makes the design more realistic than a reachability-only topology.

4. REQUIREMENTS ANALYSIS

4.1 Functional Requirements

ID	Requirement	Validation Method
FR-1	HQ, BR1, BR2 and BR3 must be logically separated into unique site blocks.	Address plan and routing table review
FR-2	Branches must reach HQ over encrypted VPN tunnels.	IKE/IPsec SA state and endpoint ping tests
FR-3	Branch-to-branch communication must be possible through HQ.	BR1-BR2, BR1-BR3 and BR2-BR3 ping tests
FR-4	DMZ web services must be separated from the inside network.	DMZ interface, VLAN and ACL checks
FR-5	Remote worker access must use a separate VPN pool.	Remote access network documentation
FR-6	Centralized logging must be available conceptually through HQ-SYSLOG-SRV.	Device/syslog architecture review

4.2 Security Requirements

ID	Security Requirement	Design Response
SR-1	Guest devices must not reach internal 10.x.x.x corporate networks.	Guest ACL denies internal private banking blocks.
SR-2	ATM devices must be restricted to required banking services.	ATM VLAN ACL and default-deny behavior.
SR-3	DMZ services must not be placed in the same trust zone as internal users.	HQ-ASA DMZ interface and DMZ VLAN.
SR-4	Branch traffic must be encrypted over the WAN.	Site-to-site IPsec VPN tunnels.
SR-5	Firewall policy must prefer explicit permit and implicit deny.	ASA ACLs and segment ACLs.
SR-6	Management traffic must be separated from user traffic.	VLAN 99 management network.

4.3 Non-Functional Requirements

The network must be understandable, scalable and demonstrable in a lab environment. The design should allow additional branches and VLANs to be added without rewriting the whole address plan. It should also be possible to troubleshoot failures by using standard network commands such as show interface, show ip route, show ospf neighbor, show crypto ikev1 sa, show crypto ipsec sa, show access-list, packet-tracer, ping and traceroute.

5. FINAL EVE-NG TOPOLOGY

The final topology contains four major locations and one remote worker area. All locations are connected to a central Internet/WAN cloud. The HQ site is the hub; branch sites connect to HQ with encrypted tunnels. The provided final topology photograph shows the overall organization: Main Retail Branch on the upper-left, Headquarters on the lower-left, Satellite Branch 1 and Satellite Branch 2 on the right side, and Remote Worker near the central WAN cloud.

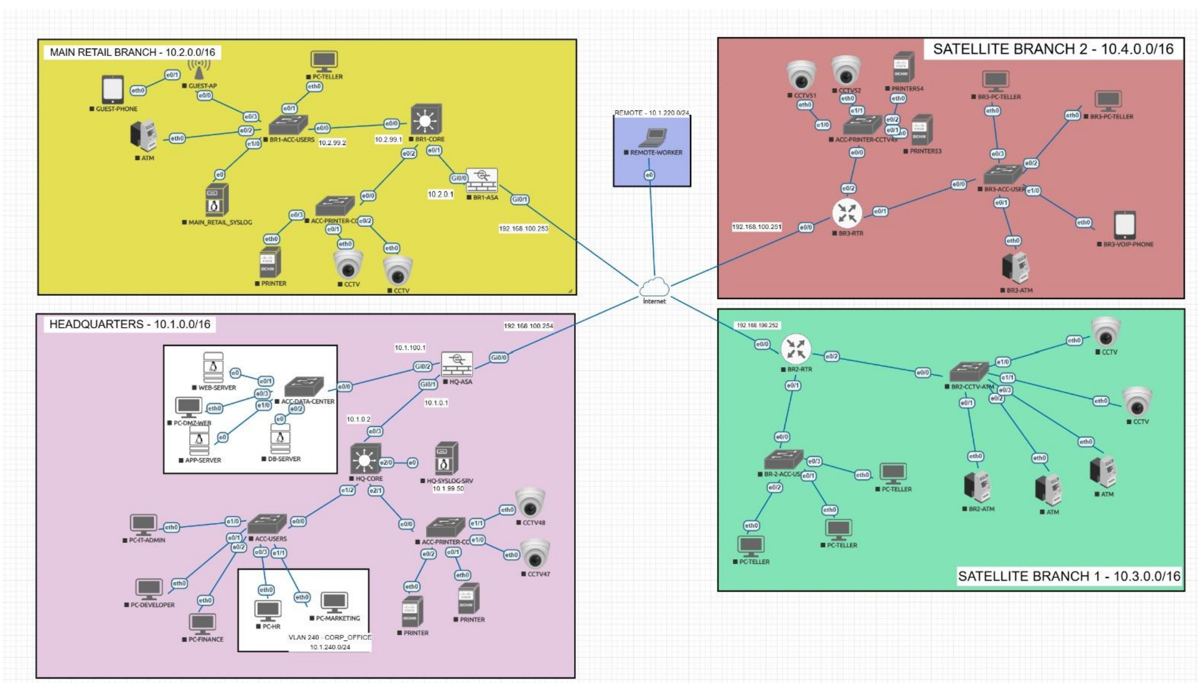


Figure 5.1. Final EVE-NG topology photograph showing HQ, Main Retail Branch, Satellite Branch 1, Satellite Branch 2 and Remote Worker areas.

5.1 Logical Topology Interpretation

Figure 5.2 presents a cleaned logical view of the final topology. It is not a replacement for the EVE-NG screenshot; rather, it summarizes the network roles and site-to-site relationship in a readable form for the report.

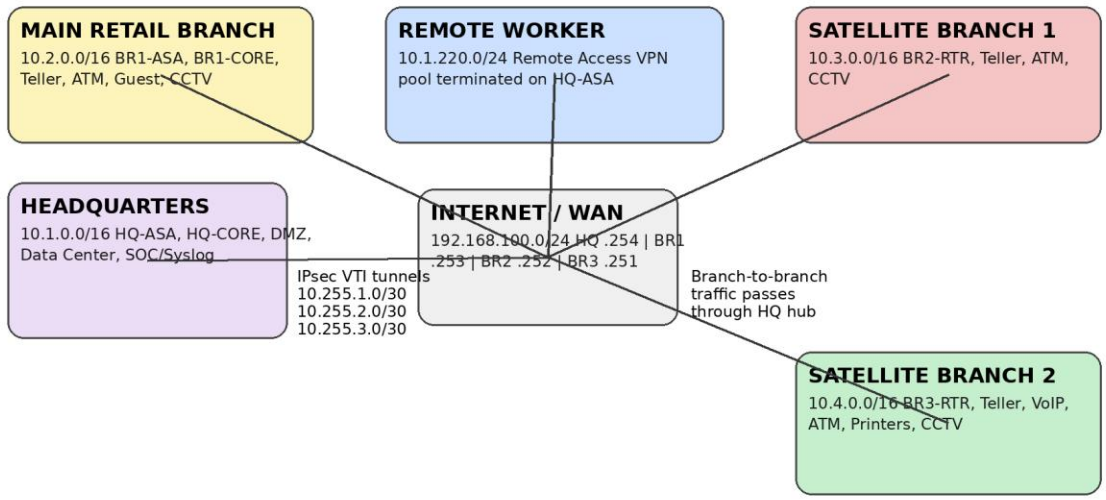


Figure 5.2. Logical hub-and-spoke interpretation of the final FKG Bank topology.

5.2 Site Summary

Location	Label	IP Block	Purpose
Headquarters	HQ	10.1.0.0/16	Core banking services, DMZ, user access, syslog/SOC and VPN hub
Main Retail Branch	BR1	10.2.0.0/16	Main retail branch with ASA, core routing, teller, ATM, guest and CCTV services
Satellite Branch 1	BR2	10.3.0.0/16	Satellite branch with router-based connectivity, teller/ATM/CCTV segments
Satellite Branch 2	BR3	10.4.0.0/16	Satellite branch with router-based connectivity, teller, VoIP, ATM, printers and CCTV
Remote Worker	Remote	10.1.220.0/24	Remote access VPN pool for staff access to banking resources

6. IP ADDRESSING AND VLSM STRATEGY

The final addressing scheme uses the private 10.0.0.0/8 space for enterprise sites and the 192.168.100.0/24 space as the simulated Internet/WAN transport segment. Each location receives a unique second-octet site identifier. This makes route interpretation and troubleshooting easier: 10.1.x.x belongs to HQ, 10.2.x.x belongs to Main Retail Branch, 10.3.x.x belongs to Satellite Branch 1 and 10.4.x.x belongs to Satellite Branch 2.

Network	Role	Example Gateway / Device	Notes
10.1.0.0/16	Headquarters	HQ-CORE / HQ-ASA inside	HQ users, DMZ, servers, SOC and remote access termination
10.2.0.0/16	Main Retail Branch	BR1-CORE / BR1-ASA	Main branch endpoint VLANs and local services
10.3.0.0/16	Satellite Branch 1	BR2-RTR	Branch 1 teller, ATM and CCTV endpoints
10.4.0.0/16	Satellite Branch 2	BR3-RTR	Branch 2 teller, ATM, voice, printers and CCTV endpoints
10.1.220.0/24	Remote Worker VPN	HQ-ASA VPN pool	Remote access staff network
10.255.0.0/24	VPN tunnel transit	VTI point-to-point links	IPsec VTI addressing
192.168.100.0/24	WAN/Internet cloud	HQ .254, BR1 .253, BR2 .252, BR3 .251	Simulated public/private WAN segment

6.1 WAN and VPN Transit Plan

Connection	WAN Peer	Tunnel Network	HQ Tunnel IP	Branch Tunnel IP
HQ-BR1	192.168.100.254 <-> 192.168.100.253	10.255.1.0/30	10.255.1.1	10.255.1.2
HQ-BR2	192.168.100.254 <-> 192.168.100.252	10.255.2.0/30	10.255.2.1	10.255.2.2
HQ-BR3	192.168.100.254 <-> 192.168.100.251	10.255.3.0/30	10.255.3.1	10.255.3.2

The tunnel networks are separated from endpoint VLANs. This separation allowed the team to diagnose tunnel-source traffic separately from endpoint-source traffic. For example, BR3 tunnel-source pings required explicit tunnel network permissions on HQ-ASA before the tests succeeded.

7. VLAN AND ZONE DESIGN

The VLAN design is role-based. VLAN IDs are reused consistently where possible: VLAN 50 for ATM, VLAN 60 for Teller, VLAN 99 for Management, VLAN 300 for Voice, VLAN 310 for Printers, VLAN 320 for CCTV and VLAN 400 for Guest. The subnet changes according to the site block.

7.1 Active and Reserved VLANs

The presentation includes an extended production-style VLAN catalog. The live EVE-NG topology implements a practical subset of that catalog. In the report tables, active endpoint VLANs are documented alongside reserve VLANs that represent future production expansion. This avoids the mistake of claiming that every VLAN listed in the design deck is fully populated in the final lab.

7.2 Headquarters VLAN Catalog

VLAN	Name	Subnet	Zone	Implementation Status
99	MGMT_NETWORK	10.1.99.0/24	Management	Design reserve / device management
100	DMZ_PUBLIC	10.1.100.0/24	DMZ	Active: WEB-SERVER and PC-DMZ-WEB
110	APP_SERVERS	10.1.110.0/24	Application/SOC	Active: HQ-SYSLOG-SRV concept at 10.1.110.20
120	DB_SERVERS	10.1.120.0/24	Database	Protected server zone
200	IT_ADMIN	10.1.200.0/24	Admin	Active: PC-IT-ADMIN
210	DEVELOPERS	10.1.210.0/24	Developer	Active: PC-DEVELOPER
230	FINANCE_ACC	10.1.230.0/24	Finance	Active: PC-FINANCE
240	CORP_OFFICE	10.1.240.0/24	Corporate office	Active: HR, Marketing and office clients
310	PRINTERS	10.1.31.0/24	Printer	Active: printer devices
320	CCTV_SEC	10.1.32.0/24	CCTV	Active: CCTV46 and CCTV47

7.3 Branch VLAN Catalog

VLAN	Name	Subnet Pattern	Purpose
50	ATM_KIOSK	10.x.50.0/24	ATM devices and kiosks
60	TELLER_OPS	10.x.60.0/24	Teller and cashier workstations
99	MGMT_NETWORK	10.x.99.0/24	Management of branch routers/switches/APs
240	CORP_OFFICE	10.x.240.0/24	Corporate office users where present
300	VOICE_VOIP	10.x.30.0/24	VoIP phones
310	PRINTERS	10.x.31.0/24	Printers and print servers
320	CCTV_SEC	10.x.32.0/24	CCTV cameras and local recording devices
400	GUEST_WIFI	192.168.x0.0/24	Guest wireless internet, isolated from 10.0.0.0/8

8. HEADQUARTERS ARCHITECTURE

Headquarters is the central control point of the FKG Bank network. It contains the HQ-ASA firewall, HQ-CORE switch, data center access switch, user access switch, printer/CCTV access switch, DMZ server area and central monitoring/SOC node. It uses the 10.1.0.0/16 block and acts as the hub for site-to-site VPN connectivity.

8.1 HQ Device Inventory

Device	Role	Important IP / Interface	Notes
HQ-ASA	Perimeter firewall and VPN hub	WAN 192.168.100.254, DMZ 10.1.100.1, inside 10.1.0.1	Terminates branch VPN tunnels and remote access VPN concept
HQ-CORE	Layer 3/core switch	10.1.0.1 / 10.1.0.2	Core routing and uplink role
ACC-USERS	User access switch	Access ports for IT, developer, HR, finance, marketing	Connects HQ user endpoints
ACC-DATA-CENTER	Data center/DMZ access switch	DMZ ports in VLAN 100	Connects WEB-SERVER, PC-DMZ-WEB and server area
ACC-PRINTER-CC	Printer/CCTV access switch	Printer/CCTV interfaces	Connects printers and CCTV cameras
HQ-SYSLOG-SRV	Central syslog/SOC server	10.1.110.20 concept	Collects ASA and network event logs

8.2 HQ Security Role

HQ-ASA is the main policy decision point. It controls WAN-facing access, branch tunnel termination, DMZ isolation and remote access VPN entry. The design relies on explicit firewall rules, implicit deny behavior and NAT exemption for private-to-private tunnel traffic. The internal HQ network is not treated as fully trusted. Instead, traffic between zones is controlled through VLAN and firewall policies.

8.3 DMZ Design

The DMZ uses 10.1.100.0/24. The ASA DMZ gateway is 10.1.100.1. The Web Server and PC-DMZ-WEB are placed in this zone. During implementation, DMZ local connectivity failed because the PC-DMZ-WEB and Web Server had the same IP address. This created an ARP conflict. The conflict was resolved by assigning unique addresses, such as 10.1.100.10 for PC-DMZ-WEB and 10.1.100.20 for WEB-SERVER, and then clearing ARP state on ASA.

A protected database server should not be treated as a public DMZ host in a production design. The slide deck separates DB_SERVERS into 10.1.120.0/24. Therefore, the report documents DB-SERVER as a protected server/data center component rather than an Internet-facing DMZ service. This is more consistent with a secure three-tier banking service model.

9. MAIN RETAIL BRANCH ARCHITECTURE

The Main Retail Branch is the largest branch site in the final topology and uses 10.2.0.0/16. It contains BR1-ASA, BR1-CORE, BR1-ACC-USERS, ACC-PRINTER-CC, a syslog server, teller workstation, ATM, guest phone, guest AP, printer and CCTV devices. The branch uses an ASA-based security boundary and a core routing device for local VLAN routing.

9.1 BR1 Device Roles

Device	Role	Notes
BR1-ASA	Branch firewall / VPN endpoint	WAN 192.168.100.253; inside link to BR1-CORE
BR1-CORE	Branch core router/switch	Gateway for branch VLANs and local policies
BR1-ACC-USERS	Access switch	Connects PC-TELLER, ATM and guest-side devices
ACC-PRINTER-CC	Printer/CCTV access	Connects printer and CCTV cameras
MAIN_RETAIL_SYSLOG	Branch syslog server	Local logging concept
GUEST-AP	Guest wireless access point	Guest network access via isolated VLAN

9.2 BR1 VLANs

VLAN	Name	Subnet	Purpose / Policy
50	ATM_KIOSK	10.2.50.0/24	Restricted ATM network; direct internal access is limited
60	TELLER_OPS	10.2.60.0/24	Teller workstation network
99	MGMT_NETWORK	10.2.99.0/24	Network device management
240	CORP_OFFICE	10.2.240.0/24	Branch office/manager users, design capacity
300	VOICE_VOIP	10.2.30.0/24	Branch IP phones
310	PRINTERS	10.2.31.0/24	Printer network
320	CCTV_SEC	10.2.32.0/24	CCTV network
400	GUEST_WIFI	192.168.20.0/24	Guest devices, denied to internal 10.0.0.0/8

9.3 BR1 Security Behavior

BR1 has stricter behavior than router-based branches because ASA enforces tunnel ACLs and security policies. During branch-to-branch testing, traffic from BR2 and BR3 to BR1 was initially blocked because BR1-ASA only permitted HQ_INTERNAL to

BR1_INTERNAL. The issue was corrected by adding BR2_INTERNAL and BR3_INTERNAL permissions to the VTI inbound ACL. This demonstrated why ASA-based branches require both routing and firewall policy checks.

10. SATELLITE BRANCH ARCHITECTURES

Satellite Branch 1 and Satellite Branch 2 are smaller branches connected to HQ through router-based site-to-site VPN. They use the same hub-and-spoke model as BR1, but their local device sets are smaller and more focused on teller, ATM, CCTV, voice and printer services.

10.1 Satellite Branch 1 - 10.3.0.0/16

Device	Role	Important Details
BR2-RTR	Branch router / VPN endpoint	WAN 192.168.100.252; router-on-a-stick VLAN gateways
BR2-CCTV-ACC	CCTV access switch	Connects CCTV cameras and ATM-side access
BR2-2-ACC-US	User access switch	Connects teller PCs and user-facing devices
PC-TELLER x2	Teller endpoints	Observed DHCP addresses include 10.3.60.11 and 10.3.60.12
ATM x2 + BR2-ATM	ATM endpoints	Observed DHCP addresses include 10.3.50.11 and 10.3.50.12
CCTV x3	Security cameras	CCTV VLAN 10.3.32.0/24

10.2 Satellite Branch 2 - 10.4.0.0/16

Device	Role	Important Details
BR3-RTR	Branch router / VPN endpoint	WAN 192.168.100.251; site-to-site tunnel to HQ
BR3-ACC-USER	User access switch	Connects teller, VoIP and ATM endpoints
ACC-PRINTER-CC	Printer access switch	Connects PRINTERS3 and PRINTERS4
BR3-PC-TELLER x2	Teller endpoints	10.4.60.0/24 teller network
BR3-VOIP-PHONE	Voice endpoint	10.4.30.0/24 voice network
BR3-ATM	ATM endpoint	10.4.50.0/24 ATM network
CCTV S1/S2	Security cameras	10.4.32.0/24 CCTV network

10.3 Correction from Earlier Drafts

The final report uses 10.4.0.0/16 for Satellite Branch 2. Any earlier draft values based on 10.5.0.0/16 were implementation-stage notes and are not used in this final document. Tunnel addressing remains in 10.255.3.0/30, but the internal BR3 site block is 10.4.0.0/16.

11. REMOTE WORKER VPN DESIGN

The Remote Worker segment represents a bank employee working from outside the corporate network. The remote worker uses a secure VPN connection to HQ-ASA and receives an address from 10.1.220.0/24. This prevents the remote device from being placed directly into the same subnet as internal HQ users. It also allows ACLs and logs to distinguish remote-user traffic from on-site user traffic.

11.1 Remote Access Principles

- Remote workers should authenticate before receiving corporate access.
- The assigned VPN pool should be separate from normal inside user VLANs.
- Access should be limited to required corporate services.
- Split tunneling can be used so only corporate traffic uses the VPN path.
- NAT traversal support is necessary for users behind home routers or carrier NAT.

11.2 Security Justification

Remote access is a high-risk entry point because users connect from unmanaged networks. The design therefore uses a separate VPN pool, centralized termination on HQ-ASA and policy-based access. In future work, the model can be strengthened with multi-factor authentication, endpoint posture checks and RADIUS/TACACS+ integration.

12. ROUTING, SWITCHING AND VPN DESIGN

12.1 OSPF Routing Strategy

OSPF is used as the internal dynamic routing protocol. The HQ and VPN tunnel networks form the central transit path. Branch internal networks are advertised toward HQ over the IPsec VTI tunnels. HQ then learns branch routes dynamically and can forward traffic between branches through the hub. OSPF was preferred over purely static routes because it scales better when additional VLANs and branches are introduced.

Device	Advertised Networks	OSPF Role
HQ-ASA	10.1.0.0/16 and 10.255.x.0/30 tunnel networks	Central hub and route exchange point
BR1-ASA	10.2.0.0/16 and 10.255.1.0/30	Main branch route participant
BR2-RTR	10.3.0.0/16 and 10.255.2.0/30	Satellite branch 1 route participant
BR3-RTR	10.4.0.0/16 and 10.255.3.0/30	Satellite branch 2 route participant

12.2 802.1Q Trunking and Access Port Model

Branch routers use subinterfaces where multiple VLANs must be carried over a single link. Switch uplinks are configured as 802.1Q trunks, while endpoint ports are configured as access ports in the appropriate VLAN. This model allows multiple logical networks to share physical links without collapsing their broadcast domains.

12.3 Site-to-Site IPsec VPN

Each branch has an independent VPN tunnel to HQ. The implementation follows a hub-and-spoke model. Branch-to-branch traffic is not sent directly between branches; it is routed through HQ so that the central ASA and routing policy can inspect and control the flow. This is appropriate for a bank because centralized policy enforcement is preferred over uncontrolled direct branch mesh behavior.

12.4 NAT Exemption and Route Lookup

Private-to-private VPN traffic should not be translated like Internet-bound traffic. Therefore, NAT exemption rules are required on ASA. During testing, packet-tracer showed some HQ-to-branch flows with Action allow but output-interface outside. This indicated that the packet was permitted but selected the wrong egress interface because NAT did not perform route lookup. The issue was corrected by adding no-proxy-arp route-lookup to the relevant NAT exemption rules. After the correction, packet-tracer showed VTI_BR2 and VTI_BR3 as the output interfaces.

12.5 CEF Behavior in EVE-NG

BR2 and BR3 router-based VPN paths required no ip cef in the EVE-NG IOS images. This is not presented as a production recommendation. It is documented as a lab-specific workaround for transit forwarding issues in the emulator. On real Cisco routers, CEF is normally kept enabled for performance. In this lab, routing, IPsec and OSPF were correct, but endpoint transit traffic failed until CEF was disabled. This distinction is important for defense during project presentation.

13. SECURITY ARCHITECTURE

The security architecture follows defense in depth. Perimeter security, VPN encryption, VLAN segmentation, ACL filtering, DMZ separation, management isolation and logging are treated as complementary controls. No single control is expected to protect the whole system alone.

13.1 Zero Trust Interpretation

Zero Trust in this project means that location alone does not imply trust. A device inside a branch is not automatically allowed to reach all internal resources. A guest phone is physically in a branch but is not trusted. An ATM device belongs to the bank but is still restricted. A DMZ server may be owned by the bank but is treated as exposed because it is reachable from outside.

13.2 Least Privilege Policy Matrix

Source Zone	Allowed Targets	Denied / Restricted Targets	Reason
Teller VLAN	HQ application/service zones and required branch resources	ATM management, CCTV management, guest networks	Teller devices need operational access, not full network control
ATM VLAN	Required banking/service subnet and local gateway	General user VLANs, guest networks, arbitrary branch-to-branch targets	ATM integrity and transaction isolation
Guest VLAN	Internet path if available	All 10.0.0.0/8 internal networks	Guest devices are untrusted
DMZ_PUBLIC	Specific inbound HTTP/HTTPS and controlled return traffic	Unrestricted inside access	Public services must be isolated
Management VLAN	Network device management interfaces	Normal user/guest flows	Administrative control-plane separation

13.3 ASA Security Policies

ASA policies are used at HQ and BR1. HQ-ASA controls outside, inside, DMZ and VTI interfaces. BR1-ASA controls the Main Retail Branch VPN edge and branch ingress policies. The key principle is explicit permission for legitimate flows and implicit denial for everything else. This was visible in tests where administratively prohibited messages occurred for intentionally blocked segments.

13.4 Guest Isolation

Guest isolation was tested directly. When the guest phone could reach BR2 and BR3 teller networks, the policy was identified as incomplete. The correct policy is to deny guest access to all internal 10.0.0.0/8 corporate networks while allowing only Internet-bound traffic in a real deployment. This negative test is a strength of the project because it shows that the team did not simply accept successful pings as success.

14. MONITORING AND SOC-ORIENTED LOGGING

The project includes a centralized logging concept through HQ-SYSLOG-SRV. The presentation describes this node as the central collection point for firewall events, VPN status changes, denied packets and audit-relevant records. In a real production environment, the syslog server would be integrated with a SIEM platform, alerting rules, correlation logic and retention policies.

14.1 Events to Monitor

- VPN tunnel establishment and teardown events.
- Failed IKE negotiation attempts and authentication errors.
- ACL deny events for guest, ATM and DMZ policies.
- Interface up/down events on WAN, tunnel and trunk links.
- Configuration changes on ASA, routers and switches.
- Repeated ICMP or port scanning attempts against DMZ services.

14.2 SOC Readiness

The current lab does not implement a complete SOC with dashboards and incident response automation. However, it places the necessary network component - centralized logging - into the architecture. This is a realistic step because security controls that cannot be monitored are difficult to defend during audit and incident response.

15. IMPLEMENTATION AND CONFIGURATION SUMMARY

This section summarizes the implemented configuration logic. The full representative configuration appendix is provided at the end of the report. The appendix is normalized for readability and final report consistency. It should be compared with device show running-config outputs before submission if exact command-level matching is required.

15.1 HQ-ASA Summary

- Outside/WAN interface: 192.168.100.254.
- Inside/core interface: 10.1.0.1 / HQ inside role.
- DMZ interface: 10.1.100.1.
- IPsec VTI tunnels to BR1, BR2 and BR3 using 10.255.1.0/30, 10.255.2.0/30 and 10.255.3.0/30.
- Objects for HQ_INTERNAL, BR1_INTERNAL, BR2_INTERNAL, BR3_INTERNAL and tunnel networks.
- NAT exemption with route-lookup for private VPN traffic.
- ACL rules for HQ, branch, DMZ and tunnel traffic.

15.2 Branch Summary

- BR1 uses ASA at the branch edge and BR1-CORE for branch VLAN routing.
- BR2 uses BR2-RTR as the WAN/VPN router and VLAN gateway.
- BR3 uses BR3-RTR as the WAN/VPN router and VLAN gateway.
- BR2 and BR3 required no ip cef as an EVE-NG IOS workaround.
- Branch access switches use access VLANs for endpoints and trunk links toward branch routers or core devices.

15.3 Endpoint Addressing Observations

Some endpoints received DHCP-style addresses rather than fixed .10 addresses. For example, BR2 teller devices were observed at 10.3.60.11 and 10.3.60.12, and BR2 ATM devices at 10.3.50.11 and 10.3.50.12. This explains why initial tests toward 10.3.60.10 and 10.3.50.10 failed. The report now distinguishes expected subnet/gateway behavior from exact endpoint host numbering.

16. VALIDATION AND TEST RESULTS

The project was validated through both infrastructure-level and endpoint-level tests. This was necessary because tunnel status alone does not prove endpoint reachability. Similarly, endpoint ping success alone does not prove that restricted networks are correctly isolated. The validation plan therefore contains multiple layers.

16.1 Infrastructure Validation

Validation Item	Expected Result	Observed / Final Result
HQ-ASA interface status	WAN, inside, DMZ and tunnel interfaces up where used	Confirmed during implementation
IKE/IPsec tunnel state	Active SAs for BR1, BR2 and BR3	Confirmed; IPsec counters increased during tests
OSPF neighbor state	FULL neighbor relationships over tunnel links	Confirmed for branch tunnels
Routing table	HQ learns 10.2/10.3/10.4 branch networks	Confirmed after OSPF and NAT fixes
Switch VLAN membership	Endpoint ports in correct access VLANs	Confirmed through show vlan brief and local gateway tests

16.2 Positive Connectivity Tests

Test Category	Representative Tests	Final Outcome
HQ to BR1	HQ-CORE to BR1 teller network	Successful; ATM intentionally restricted depending on ACL
HQ to BR2	HQ-CORE to BR2 teller/ATM actual DHCP addresses	Successful after NAT route-lookup and no ip cef fixes
HQ to BR3	HQ-CORE to 10.4.x branch endpoints	Successful after final addressing correction
Branch to HQ	BR1/BR2/BR3 endpoints to HQ core/service networks	Successful for authorized endpoints
Branch to Branch	BR1-BR2, BR1-BR3, BR2-BR3 teller flows through HQ	Successful after BR1-ASA ACL correction
DMZ local	ASA to PC-DMZ-WEB and WEB-SERVER	Successful after duplicate IP correction

16.3 Negative Security Tests

Restricted Flow	Expected Result	Interpretation
Guest to internal corporate 10.x.x.x networks	Administratively prohibited or timeout	Correct; guest isolation policy
ATM to arbitrary internal networks	Administratively prohibited or restricted	Correct; ATM least-privilege model
External/unauthorized ICMP to DMZ or inside	Blocked unless explicitly allowed	Correct; stealth and implicit deny behavior
DMZ to unrestricted inside resources	Blocked	Correct; DMZ isolation

17. TROUBLESHOOTING AND LESSONS LEARNED

17.1 NAT Route-Lookup Issue on ASA

One important troubleshooting case occurred when packet-tracer showed Action allow but output-interface outside for traffic that should have gone through a VPN tunnel. This indicated that the ASA was not using the route table as expected because the NAT exemption rule lacked route-lookup. The fix was to remove the old NAT exemption and add a route-lookup version. After this, packet-tracer showed VTI_BR2 as the correct egress interface.

17.2 EVE-NG CEF Forwarding Issue

BR2 and BR3 routers could ping HQ from their VLAN source addresses, and IPsec/OSPF appeared correct, yet HQ-to-endpoint traffic failed. Disabling CEF solved the transit forwarding issue. The report documents this as a lab-image workaround, not a real-world best practice. This distinction demonstrates an understanding of the difference between emulator behavior and production router design.

17.3 BR1 ASA Branch-to-Branch ACL Issue

BR1 could reach BR2 and BR3, but BR2 and BR3 initially could not reach BR1 teller endpoints. This was caused by BR1-ASA VTI ACL entries that permitted only HQ_INTERNAL to BR1_INTERNAL. Adding BR2_INTERNAL and BR3_INTERNAL as permitted sources resolved the issue. The lesson is that stateful firewall policies must be updated when a topology evolves from HQ-only communication to branch-to-branch communication.

17.4 DMZ Duplicate IP Conflict

The PC-DMZ-WEB and WEB-SERVER were found to share the same IP address. This created an ARP conflict and prevented reliable ASA-to-DMZ reachability. Assigning unique addresses and clearing ARP resolved the issue. This case highlights why duplicate IP checks and ARP inspection are essential in Layer 2 troubleshooting.

17.5 Wrong Target IPs During Testing

BR2 endpoints used 10.3.60.11/12 and 10.3.50.11/12 rather than 10.3.60.10 and 10.3.50.10. Initial failures were therefore caused by testing nonexistent host addresses. The team corrected the test plan by first checking show ip on endpoints and then testing actual addresses.

18. DISCUSSION

The final topology demonstrates more than basic device connectivity. It models a security-aware banking network where different endpoint roles have different permissions. Teller systems are allowed to participate in business communication, ATM systems are restricted, guest systems are isolated and public web services are placed in DMZ. The use of HQ as the branch hub also centralizes control over branch-to-branch communication.

18.1 Strengths

- Realistic multi-site banking topology with HQ, branch and remote worker components.
- Use of IPsec VPN tunnels for branch communication over an untrusted WAN cloud.
- Role-based VLAN segmentation across user, ATM, CCTV, printer, voice, guest and management zones.
- Security policies validated through both successful and intentionally blocked traffic tests.
- Troubleshooting evidence showing NAT, ACL, CEF and ARP problem-solving.

18.2 Limitations

- The lab uses EVE-NG virtual images, so certain behaviors, such as the no ip cef workaround, are emulator-specific.
- The final design demonstrates security policies but does not implement a full SIEM, NAC or MFA platform.
- The remote worker model is documented conceptually; production-grade deployment would require stronger identity, endpoint posture and certificate management.
- High availability is not fully implemented for ASA, core switching or WAN providers.
- Some scalable VLANs remain design reserves rather than active endpoint networks in the final lab.

18.3 Validity of the Banking Scenario

The scenario is valid because it focuses on typical banking concerns: secure branch connectivity, ATM isolation, user segmentation, public service isolation, auditability and controlled remote access. The exact device models and number of endpoints are simplified for the lab, but the underlying architectural principles are transferable to larger enterprise environments.

19. ORIGINAL CONTRIBUTION

The project does not claim originality through the invention of new protocols. VLAN, OSPF, IPsec, ASA and Syslog are standard technologies. The original contribution is the scenario-based integration of these standard technologies into a role-based banking network and the validation of both allowed and denied traffic flows.

The network is not evaluated with the simplistic question "Can every device ping every other device?" Instead, it asks a more security-oriented question: "Can the right device reach the right service while the wrong device is blocked?" This is a stronger and more realistic criterion for a banking network.

19.1 Role-Based Network Behavior

Each endpoint type is treated as a different security actor. Teller devices require access to business services, ATM devices require restricted transaction paths, guest devices require isolation, DMZ servers require public service exposure without inside trust and management interfaces require administrative separation. This mapping from business role to network behavior is the core value of the project.

19.2 Negative Testing as a Design Element

A notable contribution is the inclusion of negative tests as expected outcomes. When ATM or guest traffic is blocked, the result is not automatically considered a failure. Instead, it is interpreted against the policy matrix. This prevents the common student-project mistake of treating universal ping success as the only sign of a working network.

19.3 Troubleshooting Documentation

The project also contributes by documenting realistic troubleshooting cases: NAT route-lookup mistakes, VTI ACL gaps, EVE-NG router forwarding quirks, wrong endpoint IP assumptions and DMZ ARP conflicts. These records make the report useful as an educational case study, not only as a final topology description.

20. FUTURE WORK

Future work should bring the lab closer to production banking requirements. The current topology demonstrates the core logic, but real banks require stronger redundancy, monitoring, identity integration and operational automation.

20.1 High Availability and Redundancy

HQ-ASA, HQ-CORE and WAN connectivity should be made redundant. ASA failover pairs, dual core switches, redundant uplinks and a second ISP path would reduce single points of failure. OSPF failover behavior could then be tested under link failure scenarios.

20.2 SOC and SIEM Integration

HQ-SYSLOG-SRV can be extended into a full SOC platform by integrating log parsing, alert rules, dashboards and incident tickets. Security events from ASA, routers, switches and servers could be correlated in a SIEM such as Wazuh, Splunk or Elastic Stack.

20.3 Identity and Access Control

AAA can be added through RADIUS or TACACS+. Remote VPN should be strengthened with certificate-based authentication and multi-factor authentication. Network device administration should be tied to individual admin accounts rather than shared local passwords.

20.4 IDS/IPS and Web Security

DMZ traffic could be monitored with IDS/IPS sensors. Web services could be protected with a reverse proxy or web application firewall. HTTP/HTTPS tests could be expanded beyond ping to include service-level validation.

20.5 Automation and Compliance

Configuration backup, compliance scanning and automated test scripts can be added. This would allow the team to verify that ACLs, VLANs, VPN states and routes remain correct after changes.

21. CONCLUSION

The FKG Bank project successfully models a secure multi-site banking network in EVE-NG. It includes HQ, Main Retail Branch, Satellite Branch 1, Satellite Branch 2 and Remote Worker components. The design combines IP addressing, VLAN segmentation, IPsec VPN, OSPF routing, ASA security policy, DMZ isolation and centralized logging concepts.

The final tests demonstrated that authorized branch and HQ communication works, branch-to-branch traffic can pass through HQ, and restricted segments such as Guest and ATM can be controlled by policy. The most important lesson is that a secure banking network is not measured only by connectivity. It is measured by controlled connectivity: required flows must work, unnecessary flows must be blocked and all important events must be observable.

The revised report corrects earlier inconsistencies by using 10.4.0.0/16 for Satellite Branch 2, aligning the device inventory with the final EVE-NG topology, separating active VLANs from design-reserve VLANs, incorporating the project presentation concepts and integrating the provided reference materials. The resulting report is a complete documentation artifact for the final project submission.

REFERENCES

Panta, P., n.d., Design and Implementation of a Secure, Multi-Protocol Three-Tier Enterprise Network Architecture, Softwarica College of IT and E-commerce / Coventry University, 8p.

Rosu, S.M., Popescu, M.M., Dragoi, G. and Guica, I.R., 2012, The Virtual Enterprise Network based on IPSec VPN Solutions and Management, International Journal of Advanced Computer Science and Applications, 3(11):26-34 pp.

Bakkre, M.A.J.B., n.d., Design and Simulation of a Banking Network System, ResearchGate publication, URL provided in TEZ_KAYNAK/LINKLER.txt.

Peffers, K., Tuunanen, T., Rothenberger, M.A. and Chatterjee, S., 2007, A Design Science Research Methodology for Information Systems Research, Journal of Management Information Systems, 24(3):45-77 pp.

Kent, S. and Seo, K., 2005, Security Architecture for the Internet Protocol, RFC 4301, Internet Engineering Task Force.

Kaufman, C., Hoffman, P., Nir, Y., Eronen, P. and Kivinen, T., 2014, Internet Key Exchange Protocol Version 2 (IKEv2), RFC 7296, Internet Engineering Task Force.

Moy, J., 1998, OSPF Version 2, RFC 2328, Internet Engineering Task Force.

IEEE, 2018, IEEE Standard for Local and Metropolitan Area Networks - Bridges and Bridged Networks, IEEE 802.1Q.

National Institute of Standards and Technology, 2020, Zero Trust Architecture, NIST Special Publication 800-207, Gaithersburg, MD.

National Institute of Standards and Technology, 2009, Guidelines on Firewalls and Firewall Policy, NIST Special Publication 800-41 Revision 1, Gaithersburg, MD.

National Institute of Standards and Technology, 2006, Guide to Computer Security Log Management, NIST Special Publication 800-92, Gaithersburg, MD.

Cisco Systems, n.d., Cisco ASA Series VPN and Firewall Configuration Guides, Cisco documentation portal.

Cisco Systems, n.d., OSPF Design and Configuration Guides, Cisco documentation portal.

EGE University, 2019, Thesis Writing Guide, Faculty of Engineering / Department of Computer Engineering format reference, uploaded as LisansTezYazim (3).doc.

APPENDIX A - FINAL DEVICE INVENTORY

A.1 Headquarters Devices

Category	Devices
Security / Routing	HQ-ASA; HQ-CORE
Access Layer	ACC-USERS; ACC-DATA-CENTER; ACC-PRINTER-CC
Servers	WEB-SERVER; DB-SERVER; PC-DMZ-WEB; HQ-SYSLOG-SRV
Clients	PC-IT-ADMIN; PC-DEVELOPER; PC-FINANCE; PC-HR; PC-MARKETING
Other	PRINTER x2; CCTV46; CCTV47

A.2 Main Retail Branch Devices

Category	Devices
Security / Routing	BR1-ASA; BR1-CORE
Access Layer	BR1-ACC-USERS; ACC-PRINTER-CC
Servers	MAIN_RETAIL_SYSLOG
Clients	PC-TELLER; ATM; GUEST-PHONE
Other	GUEST-AP; PRINTER; CCTV x2

A.3 Satellite Branch 1 Devices

Category	Devices
Routing	BR2-RTR
Access Layer	BR2-CCTV-ACC; BR2-2-ACC-US
Clients	PC-TELLER x2; ATM x2; BR2-ATM
Other	CCTV x3

A.4 Satellite Branch 2 Devices

Category	Devices
Routing	BR3-RTR
Access Layer	BR3-ACC-USER; ACC-PRINTER-CC
Clients	BR3-PC-TELLER x2; BR3-VOIP-PHONE; BR3-ATM
Servers / Printers	PRINTERS3; PRINTERS4
Other	CCTV S1; CCTV S2

A.5 Remote Worker

Device	Network	Purpose
REMOTE-WORKER	10.1.220.0/24	Remote access VPN user

APPENDIX B - REPRESENTATIVE CONFIGURATION APPENDIX

The following configuration blocks are normalized representative configurations based on the final report design. They are intended for documentation and review. Before final submission, exact show running-config outputs can be appended if required by the instructor.

HQ-ASA - Core Interfaces, Objects and VPN Summary

```
HQ-ASA(config)# sh running-config
: Saved
:
: Serial Number: 9ABXU5TEWDL
: Hardware:   ASAv, 2048 MB RAM, CPU Xeon 4100/6100/8100 series 2800 MHz
:
ASA Version 9.23(1)
!
hostname HQ-ASA
enable password ***** pbkdf2
service-module 0 keepalive-timeout 4
service-module 0 keepalive-counter 6
names
no mac-address auto
ip local pool VPN_POOL 10.1.220.10-10.1.220.50 mask 255.255.255.0

!
interface GigabitEthernet0/0
 nameif outside
 security-level 0
 ip address 192.168.100.254 255.255.255.0
!
interface GigabitEthernet0/1
 nameif inside
```

```
 security-level 100
 ip address 10.1.0.1 255.255.255.252
!
interface GigabitEthernet0/2
 nameif dmz
 security-level 50
 ip address 10.1.100.1 255.255.255.0
!
interface GigabitEthernet0/3
 shutdown
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet0/4
 shutdown
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet0/5
 shutdown
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet0/6
 shutdown
 no nameif
 no security-level
```

```
 no ip address
!
interface Management0/0
 no management-only
 shutdown
 no nameif
 no security-level
 no ip address
!
interface Tunnel1
 nameif VT1_BR1
 ip address 10.255.1.1 255.255.255.252
 ospf mtu-ignore
 tunnel source interface outside
 tunnel destination 192.168.100.253
 tunnel mode ipsec ipv4
 tunnel protection ipsec profile IPSEC_VTI_PROFILE
!
interface Tunnel2
 nameif VTI_BR2
 ip address 10.255.2.1 255.255.255.252
 ospf mtu-ignore
 tunnel source interface outside
 tunnel destination 192.168.100.252
 tunnel mode ipsec ipv4
 tunnel protection ipsec profile IPSEC_VTI_PROFILE
!
interface Tunnel3
 nameif VTI_BR3
 ip address 10.255.3.1 255.255.255.252
```

```
 ospf mtu-ignore
 tunnel source interface outside
 tunnel destination 192.168.100.251
 tunnel mode ipsec ipv4
 tunnel protection ipsec profile IPSEC_VTI_PROFILE
!
ftp mode passive
same-security-traffic permit intra-interface
no object-group-search access-control
object network INSIDE_NET
 subnet 10.1.0.0 255.255.0.0
object network GUEST_NET
 subnet 192.168.10.0 255.255.255.0
object network HQ_INTERNAL
 subnet 10.1.0.0 255.255.0.0
object network BR1_INTERNAL
 subnet 10.2.0.0 255.255.0.0
object network BR2_INTERNAL
 subnet 10.3.0.0 255.255.0.0
object network BR3_INTERNAL
 subnet 10.4.0.0 255.255.0.0
object network BR3_TUNNEL_NET
 subnet 10.255.3.0 255.255.255.252
object network DMZ_NET
 subnet 10.1.100.0 255.255.255.0
object network VPN_CLIENTS
 subnet 10.1.220.0 255.255.255.0
access-group OUTSIDE_TO_DMZ in interface outside
access-group DMZ_IN in interface dmz
access-group BR1_IN in interface VT1_BR1
```

access-group BR2_IN in interface VTI_BR2

access-group BR3_TO_HQ in interface VTI_BR3

access-list OUTSIDE_TO_DMZ extended permit tcp any host 10.1.100.10 eq www

access-list OUTSIDE_TO_DMZ extended permit tcp any host 10.1.100.10 eq https

access-list OUTSIDE_TO_DMZ extended permit icmp any host 10.1.100.10 echo

access-list DMZ_IN extended permit icmp host 10.1.100.10 any

access-list DMZ_IN extended permit udp host 10.1.100.10 any eq domain

access-list DMZ_IN extended permit tcp host 10.1.100.10 any eq www

access-list DMZ_IN extended permit tcp host 10.1.100.10 any eq https

access-list BR1_IN extended permit ip object BR1_INTERNAL object HQ_INTERNAL

access-list BR1_IN extended permit tcp object BR1_INTERNAL object DMZ_NET eq
www

access-list BR1_IN extended permit tcp object BR1_INTERNAL object DMZ_NET eq htt
ps

access-list BR1_IN extended permit icmp object BR1_INTERNAL object DMZ_NET echo

access-list BR2_IN extended permit ip object BR2_INTERNAL object DMZ_NET

access-list BR2_IN extended permit ip object BR2_INTERNAL object HQ_INTERNAL

access-list BR2_IN extended permit icmp object BR2_INTERNAL object DMZ_NET echo

access-list BR3_TO_HQ extended permit ip object BR3_INTERNAL object
HQ_INTERNAL

access-list BR3_TO_HQ extended permit ip object BR3_TUNNEL_NET object
HQ_INTERNAL

access-list BR3_TO_HQ extended permit ip object HQ_INTERNAL object
BR3_TUNNEL_NET

access-list BR3_TO_HQ extended permit ip object BR3_INTERNAL object DMZ_NET

access-list BR3_TO_HQ extended permit icmp object BR3_INTERNAL object
DMZ_NET echo

access-list BR2_VPN extended permit ip any object BR2_INTERNAL

access-list BR1_VPN extended permit ip any object BR1_INTERNAL

access-list SPLIT_TUNNEL standard permit 10.0.0.0 255.0.0.0

access-list SPLIT_TUNNEL standard permit 192.168.0.0 255.255.0.0

pager lines 23

mtu outside 1500

mtu inside 1500

mtu dmz 1500

no failover

no failover wait-disable

no monitor-interface service-module

icmp unreachable rate-limit 1 burst-size 1

icmp permit any outside

icmp permit any inside

icmp permit any dmz

no asdm history enable

arp timeout 14400

no arp permit-nonconnected

arp rate-limit 8192

logging enable

logging timestamp

logging trap informational

logging host inside 10.1.99.50

nat (any,outside) source dynamic BR1_INTERNAL interface

nat (inside,outside) source static HQ_INTERNAL HQ_INTERNAL destination static BR
1_INTERNAL BR1_INTERNAL no-proxy-arp route-lookup

nat (inside,outside) source static HQ_INTERNAL HQ_INTERNAL destination static BR
2_INTERNAL BR2_INTERNAL no-proxy-arp route-lookup

nat (inside,outside) source static HQ_INTERNAL HQ_INTERNAL destination static BR
3_INTERNAL BR3_INTERNAL no-proxy-arp route-lookup

nat (any,outside) source dynamic BR3_INTERNAL interface

nat (any,outside) source dynamic BR2_INTERNAL interface

nat (any,dmz) source static BR1_INTERNAL BR1_INTERNAL destination static
DMZ_NET
DMZ_NET no-proxy-arp route-lookup

```
nat (any,dmz) source static BR2_INTERNAL BR2_INTERNAL destination static
DMZ_NET
DMZ_NET no-proxy-arp route-lookup
nat (any,dmz) source static BR3_INTERNAL BR3_INTERNAL destination static
DMZ_NET
DMZ_NET no-proxy-arp route-lookup
nat (inside,outside) source static any any destination static VPN_CLIENTS VPN_CL
IENTS no-proxy-arp route-lookup
nat (dmz,outside) source static any any destination static VPN_CLIENTS VPN_CLIEN
TS no-proxy-arp route-lookup
!
object network INSIDE_NET
 nat (inside,outside) dynamic interface
object network GUEST_NET
 nat (inside,outside) dynamic interface
object network BR1_INTERNAL
 nat (outside,outside) dynamic interface
object network BR2_INTERNAL
 nat (outside,outside) dynamic interface
object network DMZ_NET
 nat (dmz,outside) dynamic interface
router ospf 1
 network 10.1.100.0 255.255.255.0 area 0
 network 10.1.0.0 255.255.0.0 area 0
 network 10.255.1.0 255.255.255.252 area 0
 network 10.255.2.0 255.255.255.252 area 0
 network 10.255.3.0 255.255.255.252 area 0
 log-adj-changes
!
route outside 0.0.0.0 0.0.0.0 192.168.100.1 1
route inside 10.1.0.0 255.255.0.0 10.1.0.2 1
```

```
timeout xlate 3:00:00
timeout pat-xlate 0:00:30
timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 sctp 0:02:00 icmp 0:00:02
timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00
timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00
timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute
timeout tcp-proxy-reassembly 0:01:00
timeout floating-conn 0:00:00
timeout conn-holddown 0:00:15
timeout igp stale-route 0:01:10
user-identity default-domain LOCAL
aaa authentication login-history
no snmp-server location
no snmp-server contact
crypto ipsec ikev1 transform-set MY_TRANSFORM_SET esp-aes esp-sha-hmac
crypto ipsec ikev2 ipsec-proposal VPN_PROPOSAL
 protocol esp encryption aes-256
 protocol esp integrity sha-256
crypto ipsec profile IPSEC_VTI_PROFILE
 set ikev1 transform-set MY_TRANSFORM_SET
crypto ipsec security-association pmtu-aging infinite
crypto dynamic-map DYN_MAP 10 set ikev2 ipsec-proposal VPN_PROPOSAL
crypto map OUTSIDE_MAP 65535 ipsec-isakmp dynamic DYN_MAP
crypto map OUTSIDE_MAP interface outside
crypto ca trustpoint _SmartCallHome_ServerCA
 no validation-usage
 crl configure
crypto ca trustpoint _SmartCallHome_ServerCA2
 no validation-usage
 crl configure
```

```
crypto ca trustpool policy
 auto-import
crypto ca certificate chain _SmartCallHome_ServerCA
 certificate ca 0a0142800000014523c844b500000002
  30820560 30820348 a0030201 0202100a 01428000 00014523 c844b500 00000230
  0d06092a 864886f7 0d01010b 0500304a 310b3009 06035504 06130255 53311230
  10060355 040a1309 4964656e 54727573 74312730 25060355 0403131e 4964656e
  54727573 7420436f 6d6d6572 6369616c 20526f6f 74204341 2031301e 170d3134
  30313136 31383132 32335a17 0d333430 31313631 38313232 335a304a 310b3009
  06035504 06130255 53311230 10060355 040a1309 4964656e 54727573 74312730
  25060355 0403131e 4964656e 54727573 7420436f 6d6d6572 6369616c 20526f6f
  74204341 20313082 0222300d 06092a86 4886f70d 01010105 00038202 0f003082
  020a0282 020100a7 5019de3f 993dd433 46f16f51 6182b2a9 4f8f6789 5d84d953
  dd0c28d9 d7f0ffae 95437299 f9b55d7c 8ac142e1 315074d1 810d7ccd 9b21ab43
  e2acad5e 866ef309 8a1f5a32 bda2eb94 f9e85c0a ecff98d2 af71b3b4 539f4e87
  ef92bcbd ec4f3230 884b175e 57c453c2 f602978d d9622bbf 241f628d dfc3b829
  4b49783c 93608822 fc99da36 c8c2a2d4 2c540067 356e73bf 0258f0a4 dde5b0a2
  267acae0 36a51916 f5fdb7ef ae3f40f5 6d5a04fd ce34ca24 dc74231b 5d331312
  5dc40125 f630dd02 5d9fe0d5 47bdb4eb 1ba1bb49 49d89f5b 02f38ae4 2490e462
  4f4fc1af 8b0e7417 a8d17288 6a7a0149 ccb44679 c617b1da 981e0759 fa752185
  65dd9056 cefbaba5 609dc49d f952b08b bd87f98f 2b230a23 763bf733 e1c900f3
  69f94ba2 e04ebc7e 93398407 f744707e fe075ae5 b1acd118 ccf235e5 494908ca
  56c93dfb 0f187d8b 3bc113c2 4d8fc94f 0e37e91f a10e6adf 622ecb35 0651792c
  c82538f4 fa4ba789 5c9cd2e3 0d39864a 747cd559 87c23f4e 0c5c52f4 3df75282
  f1eaa3ac fd49341a 28f34188 3a13eee8 deff991d 5fbacbe8 1ef2b950 60c031d3
  73e5efbe a0ed330b 74be2020 c4676cf0 08037a55 807f464e 96a7f41e 3ee1f6d8
  09e13364 2b63d732 5e9ff9c0 7b0f786f 97bc939a f99c1290 787a8087 15d77274
  9c557478 b1bae16e 7004ba4f a0ba68c3 7bff31f0 733d3d94 2ab10b41 0ea0fe4d
  88656b79 33b4d702 03010001 a3423040 300e0603 551d0f01 01ff0404 03020106
  300f0603 551d1301 01ff0405 30030101 ff301d06 03551d0e 04160414 ed4419c0
```

```
    d3f0068b eea47bbe 42e72654 c88e3676 300d0609 2a864886 f70d0101 0b050003
    82020100 0dae9032 f6a64b7c 44761961 1e2728cd 5e54ef25 bce30890 f929d7ae
    6808e194 0058ef2e 2e7e5352 8cb65c07 ea88ba99 8b5094d7 8280df61 090093ad
    0d14e6ce c1f23794 78b05f9c b3a273b8 8f059338 cd8d3eb0 b8fbc0cf b1f2ec2d
    2d1bccec aa9ab3aa 60821b2d 3bc3843d 578a961e 9c75b8d3 30cd6008 8390d38e
    54f14d66 c05d7403 40a3ee85 7ec21f77 9c06e8c1 a7185d52 95edc9dd 259e6dfa
    a9eda33a 34d0597b daed50f3 35bfedeb 144d31c7 60f4daf1 879ce248 e2c6c537
    fb0610fa 75596631 4729da76 9a1ce982 aeef9ab9 51f78823 9a699562 3ce55580
    36d75402 fff1b95d ced4236f d845844a 5b65ef89 0cdd14a7 20cb18a5 25b40df9
    01f0a2d2 f400c874 8ea12a48 8e65db13 c4e22517 7debbe87 5b172054 51934a53
    030bec5d ca33ed62 fd45c72f 5bdc58a0 8039e6fa d7fe1314 a6ed3d94 4a4274d4
    c3775973 cd8f46be 5538effa e89132ea 97580422 de38c3cc bc6dc933 3a6a0a69
    3fa0c8ea 728f8c63 8623bd6d 3c969e95 e0494caa a2b92a1b 9c368178 edc3e846
    e2265944 751ed975 8951cd10 849d6160 cb5df997 224d8e98 e6e37ff6 5bbbaecd
    ca4a816b 5e0bf351 e1742be9 7e27a7d9 99494ef8 a580db25 0f1c6362 8ac93367
    6b3c1083 c6addea8 cd168e8d f0073771 9ff2abfc 41f5c18b ec00375d 09e54e80
    effab15c 3806a51b 4ae1dc38 2d3cdcab 1f901ad5 4a9ceed1 706cccee f457f818
    ba846e87
  quit
crypto ca certificate chain _SmartCallHome_ServerCA2
 certificate ca 0509
    308205b7 3082039f a0030201 02020205 09300d06 092a8648 86f70d01 01050500
    3045310b 30090603 55040613 02424d31 19301706 0355040a 13105175 6f566164
    6973204c 696d6974 6564311b 30190603 55040313 1251756f 56616469 7320526f
    6f742043 41203230 1e170d30 36313132 34313832 3730305a 170d3331 31313234
    31383233 33335a30 45310b30 09060355 04061302 424d3119 30170603 55040a13
    1051756f 56616469 73204c69 6d697465 64311b30 19060355 04031312 51756f56
    61646973 20526f6f 74204341 20323082 0222300d 06092a86 4886f70d 01010105
    00038202 0f003082 020a0282 0201009a 18ca4b94 0d002daf 03298af0 0f81c8ae
    4c19851d 089fab29 4485f32f 81ad321e 9046bfa3 86261a1e fe7e1c18 3a5c9c60
```

172a3a74 8333307d 615411cb edabe0e6 d2a27ef5 6b6f18b7 0a0b2dfd e93eef0a
c6b310e9 dcc24617 f85dfda4 daff9e49 5a9ce633 e62496f7 3fba5b2b 1c7a35c2
d667feab 66508b6d 28602bef d760c3c7 93bc8d36 91f37ff8 db1113c4 9c7776c1
aeb7026a 817aa945 83e205e6 b956c194 378f4871 6322ec17 6507958a 4bdf8fc6
5a0ae5b0 e35f5e6b 11ab0cf9 85eb44e9 f80473f2 e9fe5c98 8cf573af 6bb47ecd
d45c022b 4c39e1b2 95952d42 87d7d5b3 9043b76c 13f1dedd f6c4f889 3fd175f5
92c391d5 8a88d090 ecdc6dde 89c26571 968b0d03 fd9cbf5b 16ac92db eafe797c
adebaff7 16cbdbcd 252be51f fb9a9fe2 51cc3a53 0c48e60e bdc9b476 0652e611
13857263 0304e004 362b2019 02e874a7 1fb6c956 66f07525 dc67c10e 616088b3
3ed1a8fc a3da1db0 d1b12354 df44766d ed41d8c1 b222b653 1cdf351d dca1772a
31e42df5 e5e5dbc8 e0ffe580 d70b63a0 ff33a10f ba2c1515 ea97b3d2 a2b5bef2
8c961e1a 8f1d6ca4 6137b986 7333d797 969e237d 82a44c81 e2a1d1ba 675f9507
a32711ee 16107bbc 454a4cb2 04d2abef d5fd0c51 ce506a08 31f991da 0c8f645c
03c33a8b 203f6e8d 673d3ad6 fe7d5b88 c95efbcc 61dc8b33 77d34432 35096204
921610d8 9e2747fb 3b21e3f8 eb1d5b02 03010001 a381b030 81ad300f 0603551d
130101ff 04053003 0101ff30 0b060355 1d0f0404 03020106 301d0603 551d0e04
1604141a 8462bc48 4c332504 d4eed0f6 03c41946 d1946b30 6e060355 1d230467
30658014 1a8462bc 484c3325 04d4eed0 f603c419 46d1946b a149a447 3045310b
30090603 55040613 02424d31 19301706 0355040a 13105175 6f566164 6973204c
696d6974 6564311b 30190603 55040313 1251756f 56616469 7320526f 6f742043
41203282 02050930 0d06092a 864886f7 0d010105 05000382 0201003e 0a164d9f
065ba8ae 715d2f05 2f67e613 4583c436 f6f3c026 0c0db547 645df8b4 72c946a5
03182755 89787d76 ea963480 1720dce7 83f88dfc 07b8da5f 4d2e67b2 84fdd944
fc775081 e67cb4c9 0d0b7253 f8760707 4147960c fbe08226 93558cfe 221f6065
7c5fe726 b3f73290 9850d437 7155f692 2178f795 79faf82d 26876656 3077a637
78335210 58ae3f61 8ef26ab1 ef187e4a 5963ca8d a256d5a7 2fbc561f cf39c1e2
fb0aa815 2c7d4d7a 63c66c97 443cd26f c34a170a f890d257 a21951a5 2d9741da
074fa950 da908d94 46e13ef0 94fd1000 38f53be8 40e1b46e 561a20cc 6f588ded
2e458fd6 e9933fe7 b12cdf3a d6228cdc 84bb226f d0f8e4c6 39e90488 3cc3baeb
557a6d80 9924f56c 01fbf897 b0945beb fdd26ff1 77680d35 6423acb8 55a103d1

```
    4d4219dc f8755956 a3f9a849 79f8af0e b911a07c b76aed34 d0b62662 381a870c
    f8e8fd2e d3907f07 912a1dd6 7e5c8583 99b03808 3fe95ef9 3507e4c9 626e577f
    a75095f7 bac89be6 8ea201c5 d666bf79 61f33c1c e1b9825c 5da0c3e9 d848bd19
    a2111419 6eb2861b 683e4837 1a88b75d 965e9cc7 ef276208 e291195c d2f121dd
    ba174282 97718153 31a99ff6 7d62bf72 e1a3931d cc8a265a 0938d0ce d70d8016
    b478a53a 874c8d8a a5d54697 f22c10b9 bc5422c0 01506943 9ef4b2ef 6df8ecda
    f1e3b1ef df918f54 2a0b25c1 2619c452 100565d5 8210eac2 31cd2e
  quit
crypto ikev2 policy 10
 encryption aes-256
 integrity sha256
 group 14
 prf sha256
 lifetime seconds 86400
crypto ikev2 enable outside
crypto ikev1 enable outside
crypto ikev1 policy 10
 authentication pre-share
 encryption aes
 hash sha
 group 14
 lifetime 86400
telnet timeout 10
ssh stricthostkeycheck
ssh timeout 5
ssh key-exchange group dh-group14-sha256
console timeout 0
console serial
threat-detection basic-threat
threat-detection statistics access-list
```

```
no threat-detection statistics tcp-intercept
group-policy VPN_POLICY internal
group-policy VPN_POLICY attributes
 dns-server value 10.1.110.10
 vpn-tunnel-protocol ikev2
 split-tunnel-policy tunnelspecified
 split-tunnel-network-list value SPLIT_TUNNEL
dynamic-access-policy-record DfltAccessPolicy
tunnel-group 192.168.100.253 type ipsec-l2l
tunnel-group 192.168.100.253 ipsec-attributes
 ikev1 pre-shared-key *****
tunnel-group 192.168.100.252 type ipsec-l2l
tunnel-group 192.168.100.252 ipsec-attributes
 ikev1 pre-shared-key *****
tunnel-group 192.168.100.251 type ipsec-l2l
tunnel-group 192.168.100.251 ipsec-attributes
 ikev1 pre-shared-key *****
tunnel-group FKG_VPN type remote-access
tunnel-group FKG_VPN general-attributes
 address-pool VPN_POOL
 default-group-policy VPN_POLICY
tunnel-group FKG_VPN ipsec-attributes
 ikev2 remote-authentication pre-shared-key *****
 ikev2 local-authentication pre-shared-key *****
!
class-map inspection_default
 match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
```

```
 parameters
  message-length maximum client auto
  message-length maximum 512
  no tcp-inspection
policy-map global_policy
 class inspection_default
  inspect ip-options
  inspect netbios
  inspect rtsp
  inspect sunrpc
  inspect tftp
  inspect dns preset_dns_map
  inspect ftp
  inspect h323 h225
  inspect h323 ras
  inspect rsh
  inspect esmtp
  inspect sqlnet
  inspect sip
  inspect skinny
  inspect icmp
policy-map type inspect dns migrated_dns_map_2
 parameters
  message-length maximum client auto
  message-length maximum 512
  no tcp-inspection
policy-map type inspect dns migrated_dns_map_1
 parameters
  message-length maximum client auto
  message-length maximum 512
```

no tcp-inspection

!

service-policy global_policy global

prompt hostname context

call-home reporting anonymous

call-home

profile License

destination address http https://tools.cisco.com/its/service/oddce/services/DDCEService

destination transport-method http

profile CiscoTAC-1

no active

destination address http https://tools.cisco.com/its/service/oddce/services/DDCEService

destination address email callhome@cisco.com

destination transport-method http

subscribe-to-alert-group diagnostic

subscribe-to-alert-group environment

subscribe-to-alert-group inventory periodic monthly

subscribe-to-alert-group configuration periodic monthly

subscribe-to-alert-group telemetry periodic daily

Cryptochecksum:1db6a9e4ddecf519a6d8bb66586aacd5

: end

HQ-CORE:

HQ-CORE#sh running-config

Building configuration...

Current configuration : 10209 bytes

!

! Last configuration change at 08:58:32 UTC Tue Jun 9 2026

```
!
version 17.16
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname HQ-CORE
!
boot-start-marker
boot-end-marker
!
!
logging buffered 16384
logging console critical
no aaa new-model
!
!
!
!
!
!
!
!
!
!
ip dhcp excluded-address 10.1.99.1 10.1.99.10
ip dhcp excluded-address 10.1.140.1 10.1.140.10
ip dhcp excluded-address 10.1.100.1 10.1.100.10
ip dhcp excluded-address 10.1.110.1 10.1.110.10
ip dhcp excluded-address 10.1.120.1 10.1.120.10
ip dhcp excluded-address 10.1.130.1 10.1.130.10
```

```
ip dhcp excluded-address 10.1.50.1 10.1.50.10
ip dhcp excluded-address 10.1.60.1 10.1.60.10
ip dhcp excluded-address 10.1.70.1 10.1.70.10
ip dhcp excluded-address 10.1.200.1 10.1.200.10
ip dhcp excluded-address 10.1.210.1 10.1.210.10
ip dhcp excluded-address 10.1.230.1 10.1.230.10
ip dhcp excluded-address 10.1.240.1 10.1.240.10
ip dhcp excluded-address 10.1.250.1 10.1.250.10
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 10.1.26.1 10.1.26.10
ip dhcp excluded-address 10.1.30.1 10.1.30.10
ip dhcp excluded-address 10.1.31.1 10.1.31.10
ip dhcp excluded-address 10.1.32.1 10.1.32.10
ip dhcp excluded-address 10.1.33.1 10.1.33.10
ip dhcp excluded-address 172.16.66.1 172.16.66.10
!
ip dhcp pool POOL_MGMT
 network 10.1.99.0 255.255.255.0
 default-router 10.1.99.1
 dns-server 8.8.8.8
!
ip dhcp pool POOL_DMZ
 network 10.1.100.0 255.255.255.0
 default-router 10.1.100.1
 dns-server 8.8.8.8
!
ip dhcp pool POOL_APP
 network 10.1.110.0 255.255.255.0
 default-router 10.1.110.1
!
```

```
ip dhcp pool POOL_DB
 network 10.1.120.0 255.255.255.0
 default-router 10.1.120.1
!
ip dhcp pool POOL_ATM
 network 10.1.50.0 255.255.255.0
 default-router 10.1.50.1
!
ip dhcp pool POOL_TELLER
 network 10.1.60.0 255.255.255.0
 default-router 10.1.60.1
 dns-server 8.8.8.8 1.1.1.1
!
ip dhcp pool POOL_SWIFT
 network 10.1.70.0 255.255.255.0
 default-router 10.1.70.1
!
ip dhcp pool POOL_IT_ADMIN
 network 10.1.200.0 255.255.255.0
 default-router 10.1.200.1
 dns-server 8.8.8.8 1.1.1.1
!
ip dhcp pool POOL_VIP_EXEC
 default-router 10.1.26.1
 dns-server 8.8.8.8
!
ip dhcp pool POOL_VOICE
 network 10.1.30.0 255.255.255.0
 default-router 10.1.30.1
!
```

```
ip dhcp pool POOL_PRINTERS
 network 10.1.31.0 255.255.255.0
 default-router 10.1.31.1
!
ip dhcp pool POOL_CCTV
 network 10.1.32.0 255.255.255.0
 default-router 10.1.32.1
!
ip dhcp pool POOL_IOT
 network 10.1.33.0 255.255.255.0
 default-router 10.1.33.1
!
ip dhcp pool POOL_GUEST
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.4.4
!
ip dhcp pool POOL_QUARANTINE
 network 172.16.66.0 255.255.255.0
 default-router 172.16.66.1
!
ip dhcp pool POOL_FINANCE
 network 10.1.230.0 255.255.255.0
 default-router 10.1.230.1
 dns-server 8.8.8.8 1.1.1.1
!
ip dhcp pool POOL_CORP_OFFICE
 network 10.1.240.0 255.255.255.0
 default-router 10.1.240.1
 dns-server 8.8.8.8 1.1.1.1
```

```
!
ip dhcp pool POOL_DEVELOPER
 network 10.1.210.0 255.255.255.0
 default-router 10.1.210.1
 dns-server 8.8.8.8 1.1.1.1
!
!
!
ip audit notify log
ip audit po max-events 100
ip cef
login on-success log
no ipv6 cef
!
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-67108880
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-67108880
 revocation-check none
 rsakeypair TP-self-signed-67108880
 hash sha512
!
!
crypto pki certificate chain TP-self-signed-67108880
```

certificate self-signed 01

3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030

2F312D30 2B060355 04030C24 494F532D 53656C66 2D536967 6E65642D 43657274

69666963 6174652D 36373130 38383830 301E170D 32363034 32353136 33373537

5A170D33 36303432 34313633 3735375A 302F312D 302B0603 5504030C 24494F53

2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D363731 30383838

30308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201 0A028201

010082D1 42917B63 3BAB340B D0D3EB96 9BC10173 F7163505 1039419A
368D8B90

56954F40 6EE39BDD 96864095 76B60074 A95F3753 E15A4BBE 6BE421C9
888C8C90

8A1E19D1 CE5BA417 F83D2D88 81AB4E1C DA810C81 EDB38DE5 CB6BE3E3
F8E16BF9

D16EACD6 FE34C3BB 424F3AAB 783A060D A70399D0 5A3A759E E66EFC30
B27927D7

8BD16AC6 58D7CC91 049809F5 9A3BD789 240F978E D1E97CBD 3B33A46B
B198FA81

9C514B23 A8820A79 CBF5FBB4 634FDE88 CD6EA16E 9DAD26CF FBCB3D7C
0EBA581E

F1B5CB54 6BD3FC0C 1186030E 6035412A 74F5B738 7B1760EE 94C7B060
B69F5111

16DC854B 68BE7889 4EA9C634 82BDB2A7 DCA86164 6E01A986 43BDFF36
94E6E701

18410203 010001A3 53305130 1D060355 1D0E0416 0414B9A9 047DABC3 1E9B6927

3552E890 147E6C82 230B301F 0603551D 23041830 168014B9 A9047DAB C31E9B69

273552E8 90147E6C 82230B30 0F060355 1D130101 FF040530 030101FF 300D0609

2A864886 F70D0101 0D050003 82010100 2F872510 F0B97212 2541B5F0 FC03EE32

40D65D9B 9CFBFF9F 1806886A A3BB2651 8B0D8779 AC177B8B 3DD7D230
3F1B87A1

4386CE0E 4F8606F5 163916F0 8FFDA5B5 E67A5C42 03322386 6DB2E192
ED2BEC20

1DF76D8B C843D1C0 FE81AD3C 8FF81976 468EAE8A 5D3501B2 A9D43746
24DDDC9D

```
  32593562 4F639087 ACB9221A B109E2CB B8F30AD2 152CB613 44362951
1BA2AF80
  D82753D0 D4B5A706 ACD30870 6EB1AAF5 024885A2 A625EEE0 D55068FF
09FEA71B
  AAE9D67D D73EFB69 2AA51BCF B97C864F 2E7DB151 46DFC9B6 9D5B90FE
C658262D
  5EC2DDDF 24E6BA8B 51DDF197 2878BA64 1DFD53A7 5348CD1E DBE09D0F
E7B2A237
  1BE578DD D6CF046C 47DBAD8C F6C12E9B
        quit
!
!
memory free low-watermark processor 79497
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
vlan internal allocation policy ascending
!
!
!
!
!
interface Ethernet0/0
 description PC-GUEST
 switchport access vlan 400
 switchport mode access
 spanning-tree portfast
!
```

```
interface Ethernet0/1
 description IT_ADMIN_PC_PORT
 switchport access vlan 200
 switchport mode access
 spanning-tree portfast
!
interface Ethernet0/2
 description PC-DMZ-WEB
 switchport access vlan 100
 switchport mode access
 spanning-tree portfast
!
interface Ethernet0/3
 description LINK_TO_ASA_FW
 no switchport
 ip address 10.1.0.2 255.255.255.252
!
interface Ethernet1/0
 switchport access vlan 230
 switchport mode access
 spanning-tree portfast
!
interface Ethernet1/1
 description TRUNK_TO_ACCESS_SWITCHES
 switchport access vlan 100
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 50,60,70,99,100,110,120,200,210,230,240,260,300
 switchport trunk allowed vlan add 310,320,330,400,666
 switchport mode trunk
 spanning-tree portfast trunk
```

```
!
interface Ethernet1/2
 description TRUNK_TO_ACCESS_SWITCHES
 switchport access vlan 120
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 50,60,70,99,100,110,120,200,210,230,240,260,300
 switchport trunk allowed vlan add 310,320,330,400,666
 switchport mode trunk
 spanning-tree portfast trunk
!
interface Ethernet1/3
 switchport access vlan 60
 switchport mode access
 spanning-tree portfast
!
interface Ethernet2/0
 description SYSLOG_SERVER_UBUNTU
 switchport access vlan 99
 switchport mode access
 spanning-tree portfast
!
interface Ethernet2/1
!
interface Ethernet2/2
!
interface Ethernet2/3
!
interface Ethernet3/0
!
interface Ethernet3/1
```

```
!
interface Ethernet3/2
!
interface Ethernet3/3
!
interface Vlan50
 ip address 10.1.50.1 255.255.255.0
 ip access-group ACL_PCI_ZONE_IN in
!
interface Vlan60
 ip address 10.1.60.1 255.255.255.0
 ip access-group ACL_PCI_ZONE_IN in
!
interface Vlan70
 ip address 10.1.70.1 255.255.255.0
!
interface Vlan99
 ip address 10.1.99.1 255.255.255.0
!
interface Vlan100
 ip address 10.1.100.1 255.255.255.0
!
interface Vlan110
 ip address 10.1.110.1 255.255.255.0
!
interface Vlan120
 ip address 10.1.120.1 255.255.255.0
 ip access-group ACL_DB_IN in
!
interface Vlan130
```

```
 ip address 10.1.130.1 255.255.255.0
!
interface Vlan140
 ip address 10.1.140.1 255.255.255.0
!
interface Vlan200
 ip address 10.1.200.1 255.255.255.0
!
interface Vlan210
 ip address 10.1.210.1 255.255.255.0
 ip access-group ACL_OFFICE_ZONE_IN in
!
interface Vlan230
 ip address 10.1.230.1 255.255.255.0
 ip access-group ACL_OFFICE_ZONE_IN in
!
interface Vlan240
 ip address 10.1.240.1 255.255.255.0
 ip access-group ACL_USER_ACCESS_IN in
!
interface Vlan250
 ip address 10.1.250.1 255.255.255.0
!
interface Vlan260
 description VIP_EXEC_GATEWAY
 ip address 10.1.26.1 255.255.255.0
!
interface Vlan300
 ip address 10.1.30.1 255.255.255.0
!
```

```
interface Vlan310
 ip address 10.1.31.1 255.255.255.0
!
interface Vlan320
 ip address 10.1.32.1 255.255.255.0
!
interface Vlan330
 ip address 10.1.33.1 255.255.255.0
!
interface Vlan400
 ip address 192.168.10.1 255.255.255.0
 ip access-group ACL_GUEST_IN in
!
interface Vlan666
 ip address 172.16.66.1 255.255.255.0
!
ip forward-protocol nd
ip forward-protocol udp
!
!
ip http server
ip http secure-server
ip ssh bulk-mode 131072
ip route 0.0.0.0 0.0.0.0 10.1.0.1
!
ip access-list extended ACL_DB_IN
 10 remark === DB VLAN PROTECTION ===
 10 permit ip 10.1.200.0 0.0.0.255 any
 20 permit tcp 10.1.110.0 0.0.0.255 any eq 3306
 30 permit tcp 10.1.100.0 0.0.0.255 any eq 3306
```

```
 40 permit ip any host 10.1.120.1
 50 deny ip any any log
ip access-list extended ACL_GUEST_IN
 10 remark === GUEST NETWORK ===
 10 permit udp any host 192.168.10.1 eq bootps
 20 permit udp any any eq domain
 30 deny ip any 10.1.0.0 0.0.255.255
 40 permit ip any any
ip access-list extended ACL_OFFICE_ZONE_IN
 10 remark === OFFICE RESTRICTIONS ===
 10 deny ip any 10.1.50.0 0.0.0.255 log
 20 deny ip any 10.1.60.0 0.0.0.255 log
 30 deny ip any 10.1.70.0 0.0.0.255 log
 40 permit ip any 10.1.100.0 0.0.0.255
 50 permit ip any 10.1.110.0 0.0.0.255
 60 permit ip any any
ip access-list extended ACL_PCI_ZONE_IN
 10 remark === PCI ZONE (VERY RESTRICTED) ===
 10 permit udp any any eq bootpc
 20 permit udp any any eq bootps
 30 permit ip 10.1.200.0 0.0.0.255 any
 40 permit tcp 10.1.110.0 0.0.0.255 any eq 443
 50 deny ip any any log
ip access-list extended ACL_USER_ACCESS_IN
 10 remark === USER RESTRICTIONS ===
 10 deny ip any 10.1.99.0 0.0.0.255
 20 deny ip any 10.1.50.0 0.0.0.255
 30 permit ip any any
no logging btrace
logging source-interface Vlan99
```

```
logging host 10.1.100.50
logging host 10.1.99.50
!
!
!
control-plane
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input ssh
!
!
End
```

HQ-SYSLOG-SRV

```
  GNU nano 6.2                                              rsyslog.conf
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf


#################
#### MODULES ####
#################

module(load="imuxsock") # provides support for local system logging
#module(load="immark")  # provides --MARK-- message capability

# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

# provides TCP syslog reception
#module(load="imtcp")
#input(type="imtcp" port="514")

# provides kernel logging support and enable non-kernel klog messages
module(load="imklog" permitnonkernelfacility="on")

###########################
#### GLOBAL DIRECTIVES ####
###########################

#
# Use traditional timestamp format.
# To enable high precision timestamps, comment out the following line.
#
$ActionFileDefaultTemplate RSYSLOG_TraditionalFileFormat

# Filter duplicated messages
$RepeatedMsgReduction on

#
# Set the default permissions for all log files.
#
$FileOwner syslog
$FileGroup adm
                                             [ File 'rsyslog.conf' is unwritable ]
```

```
  GNU nano 6.2                                    00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    ens3:
      dhcp4: no
      addresses:
        - 10.1.99.50/24
      routes:
        - to: default
          via: 10.1.99.1
      nameservers:
        addresses: [8.8.8.8]
  version: 2
```

Logs from other devices are sent here.

```
user@ubuntu22-server:/etc$ tail -f /var/log/syslog
Jun  9 09:26:53 10.1.0.1 %ASA-6-302015: Built inbound UDP connection 88 for outside:192.168.100.253/500 (192.168.100.253/500) to
 identity:192.168.100.254/500 (192.168.100.254/500) -1 -1
Jun  9 09:26:56 10.1.0.1 %ASA-6-602304: IPSEC: An outbound LAN-to-LAN SA (SPI= 0x19A54FF8) between 192.168.100.254 and 192.168.1
00.251 (user= 192.168.100.251) has been deleted.
Jun  9 09:26:56 10.1.0.1 %ASA-6-602304: IPSEC: An inbound LAN-to-LAN SA (SPI= 0x9A5A4D75) between 192.168.100.251 and 192.168.10
0.254 (user= 192.168.100.251) has been deleted.
Jun  9 09:27:14 10.1.0.1 %ASA-4-752012: IKEv1 was unsuccessful at setting up a tunnel.  Map Tag = __vti-crypto-map-Tunnel2-0-2.
 Map Sequence Number = 65280.
Jun  9 09:27:14 10.1.0.1 %ASA-3-752015: Tunnel Manager has failed to establish an L2L SA.  All configured IKE versions failed to
 establish the tunnel. Map Tag= __vti-crypto-map-Tunnel2-0-2.  Map Sequence Number = 65280.
Jun  9 09:27:42 10.1.0.1 %ASA-5-752004: Tunnel Manager dispatching a KEY_ACQUIRE message to IKEv1.  Map Tag = __vti-crypto-map-T
unnel2-0-2.  Map Sequence Number = 65280.
Jun  9 09:27:42 10.1.0.1 %ASA-4-752010: IKEv2 Doesn't have a proposal specified
Jun  9 09:27:42 10.1.0.1 %ASA-5-713041: IP = 192.168.100.252, IKE Initiator: New Phase 1, Intf NP Identity Ifc, IKE Peer 192.168
.100.252  local Proxy Address 0.0.0.0, remote Proxy Address 0.0.0.0,  Crypto map (__vti-crypto-map-Tunnel2-0-2)
Jun  9 09:28:14 10.1.0.1 %ASA-4-752012: IKEv1 was unsuccessful at setting up a tunnel.  Map Tag = __vti-crypto-map-Tunnel2-0-2.
 Map Sequence Number = 65280.
Jun  9 09:28:14 10.1.0.1 %ASA-3-752015: Tunnel Manager has failed to establish an L2L SA.  All configured IKE versions failed to
 establish the tunnel. Map Tag= __vti-crypto-map-Tunnel2-0-2.  Map Sequence Number = 65280.
Jun  9 09:28:42 10.1.0.1 %ASA-5-752004: Tunnel Manager dispatching a KEY_ACQUIRE message to IKEv1.  Map Tag = __vti-crypto-map-T
unnel2-0-2.  Map Sequence Number = 65280.
Jun  9 09:28:42 10.1.0.1 %ASA-4-752010: IKEv2 Doesn't have a proposal specified
Jun  9 09:28:42 10.1.0.1 %ASA-5-713041: IP = 192.168.100.252, IKE Initiator: New Phase 1, Intf NP Identity Ifc, IKE Peer 192.168
.100.252  local Proxy Address 0.0.0.0, remote Proxy Address 0.0.0.0,  Crypto map (__vti-crypto-map-Tunnel2-0-2)
Jun  9 09:28:54 10.1.0.1 %ASA-6-302016: Teardown UDP connection 88 for outside:192.168.100.253/500 to identity:192.168.100.254/5
00 duration 0:02:01 bytes 184 -1 -1
Jun  9 09:28:56 10.1.0.1 %ASA-6-302016: Teardown UDP connection 86 for outside:192.168.100.251/500 to identity:192.168.100.254/5
00 duration 0:02:07 bytes 496 -1 -1
```

BR1-ASA - Branch Edge and VTI Summary

BR1-ASA(config)# sh running-config

: Saved

:

: Serial Number: 9A1BKQ1DN7T

: Hardware: ASAv, 2048 MB RAM, CPU Xeon 4100/6100/8100 series 2800 MHz

:

ASA Version 9.23(1)

!

hostname BR1-ASA

enable password ***** pbkdf2

service-module 0 keepalive-timeout 4

service-module 0 keepalive-counter 6

names

no mac-address auto

!

```
interface GigabitEthernet0/0
 nameif inside
 security-level 100
 ip address 10.2.0.1 255.255.255.252
!
interface GigabitEthernet0/1
 nameif outside
 security-level 0
 ip address 192.168.100.253 255.255.255.0
!
interface GigabitEthernet0/2
 shutdown
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet0/3
 shutdown
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet0/4
 shutdown
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet0/5
 shutdown
```

```
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet0/6
 shutdown
 no nameif
 no security-level
 no ip address
!
interface Management0/0
 no management-only
 shutdown
 no nameif
 no security-level
 no ip address
!
interface Tunnel1
 nameif VTI_HQ
 ip address 10.255.1.2 255.255.255.252
 ospf mtu-ignore
 tunnel source interface outside
 tunnel destination 192.168.100.254
 tunnel mode ipsec ipv4
 tunnel protection ipsec profile IPSEC_VTI_PROFILE
!
ftp mode passive
no object-group-search access-control
object network OBJ_BRANCH_INTERNAL
 subnet 10.2.0.0 255.255.0.0
```

```
object network OBJ_GUEST_WIFI
 subnet 192.168.20.0 255.255.255.0
object network BR1_INTERNAL
 subnet 10.2.0.0 255.255.0.0
object network HQ_INTERNAL
 subnet 10.1.0.0 255.255.0.0
object network BR2_INTERNAL
 subnet 10.3.0.0 255.255.0.0
object network BR3_INTERNAL
 subnet 10.5.0.0 255.255.0.0
access-group BR1_TO_HQ in interface VTI_HQ
access-list BR1_TO_HQ extended permit ip object BR1_INTERNAL object
HQ_INTERNAL
access-list BR1_TO_HQ extended permit ip object HQ_INTERNAL object
BR1_INTERNAL
access-list BR1_TO_HQ extended permit ip object BR2_INTERNAL object
BR1_INTERNAL
access-list BR1_TO_HQ extended permit ip object BR3_INTERNAL object
BR1_INTERNAL
access-list VPN_ACL extended permit ip object BR1_INTERNAL object HQ_INTERNAL
access-list BR1_TOHQ extended permit ip object BR2_INTERNAL object
BR1_INTERNAL
pager lines 23
mtu inside 1500
mtu outside 1500
no failover
no failover wait-disable
no monitor-interface service-module
icmp unreachable rate-limit 1 burst-size 1
no asdm history enable
arp timeout 14400
```

```
no arp permit-nonconnected
arp rate-limit 8192
logging enable
logging timestamp
logging trap informational
nat (inside,outside) source static BR1_INTERNAL BR1_INTERNAL destination static HQ_INTERNAL HQ_INTERNAL no-proxy-arp route-lookup
!
nat (inside,outside) after-auto source dynamic OBJ_BRANCH_INTERNAL interface
nat (inside,outside) after-auto source dynamic OBJ_GUEST_WIFI interface
router ospf 1
 network 10.255.1.0 255.255.255.252 area 0
 log-adj-changes
 redistribute static subnets
!
route outside 0.0.0.0 0.0.0.0 192.168.100.1 1
route inside 10.2.30.0 255.255.255.0 10.2.0.2 1
route inside 10.2.31.0 255.255.255.0 10.2.0.2 1
route inside 10.2.32.0 255.255.255.0 10.2.0.2 1
route inside 10.2.50.0 255.255.255.0 10.2.0.2 1
route inside 10.2.60.0 255.255.255.0 10.2.0.2 1
route inside 10.2.99.0 255.255.255.0 10.2.0.2 1
route inside 10.2.240.0 255.255.255.0 10.2.0.2 1
route inside 192.168.20.0 255.255.255.0 10.2.0.2 1
timeout xlate 3:00:00
timeout pat-xlate 0:00:30
timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 sctp 0:02:00 icmp 0:00:02
timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00
timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00
timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute
```

```
timeout tcp-proxy-reassembly 0:01:00
timeout floating-conn 0:00:00
timeout conn-holddown 0:00:15
timeout igp stale-route 0:01:10
user-identity default-domain LOCAL
aaa authentication login-history
no snmp-server location
no snmp-server contact
crypto ipsec ikev1 transform-set MY_TRANSFORM_SET esp-aes esp-sha-hmac
crypto ipsec profile IPSEC_VTI_PROFILE
 set ikev1 transform-set MY_TRANSFORM_SET
crypto ipsec security-association pmtu-aging infinite
crypto ca trustpoint _SmartCallHome_ServerCA
 no validation-usage
 crl configure
crypto ca trustpoint _SmartCallHome_ServerCA2
 no validation-usage
 crl configure
crypto ca trustpool policy
 auto-import
crypto ca certificate chain _SmartCallHome_ServerCA
 certificate ca 0a0142800000014523c844b500000002
```

```
    30820560 30820348 a0030201 0202100a 01428000 00014523 c844b500 00000230
    0d06092a 864886f7 0d01010b 0500304a 310b3009 06035504 06130255 53311230
    10060355 040a1309 4964656e 54727573 74312730 25060355 0403131e 4964656e
    54727573 7420436f 6d6d6572 6369616c 20526f6f 74204341 2031301e 170d3134
    30313136 31383132 32335a17 0d333430 31313631 38313232 335a304a 310b3009
    06035504 06130255 53311230 10060355 040a1309 4964656e 54727573 74312730
    25060355 0403131e 4964656e 54727573 7420436f 6d6d6572 6369616c 20526f6f
    74204341 20313082 0222300d 06092a86 4886f70d 01010105 00038202 0f003082
```

020a0282 020100a7 5019de3f 993dd433 46f16f51 6182b2a9 4f8f6789 5d84d953
dd0c28d9 d7f0ffae 95437299 f9b55d7c 8ac142e1 315074d1 810d7ccd 9b21ab43
e2acad5e 866ef309 8a1f5a32 bda2eb94 f9e85c0a ecff98d2 af71b3b4 539f4e87
ef92bcbd ec4f3230 884b175e 57c453c2 f602978d d9622bbf 241f628d dfc3b829
4b49783c 93608822 fc99da36 c8c2a2d4 2c540067 356e73bf 0258f0a4 dde5b0a2
267acae0 36a51916 f5fdb7ef ae3f40f5 6d5a04fd ce34ca24 dc74231b 5d331312
5dc40125 f630dd02 5d9fe0d5 47bdb4eb 1ba1bb49 49d89f5b 02f38ae4 2490e462
4f4fc1af 8b0e7417 a8d17288 6a7a0149 ccb44679 c617b1da 981e0759 fa752185
65dd9056 cefbaba5 609dc49d f952b08b bd87f98f 2b230a23 763bf733 e1c900f3
69f94ba2 e04ebc7e 93398407 f744707e fe075ae5 b1acd118 ccf235e5 494908ca
56c93dfb 0f187d8b 3bc113c2 4d8fc94f 0e37e91f a10e6adf 622ecb35 0651792c
c82538f4 fa4ba789 5c9cd2e3 0d39864a 747cd559 87c23f4e 0c5c52f4 3df75282
f1eaa3ac fd49341a 28f34188 3a13eee8 deff991d 5fbacbe8 1ef2b950 60c031d3
73e5efbe a0ed330b 74be2020 c4676cf0 08037a55 807f464e 96a7f41e 3ee1f6d8
09e13364 2b63d732 5e9ff9c0 7b0f786f 97bc939a f99c1290 787a8087 15d77274
9c557478 b1bae16e 7004ba4f a0ba68c3 7bff31f0 733d3d94 2ab10b41 0ea0fe4d
88656b79 33b4d702 03010001 a3423040 300e0603 551d0f01 01ff0404 03020106
300f0603 551d1301 01ff0405 30030101 ff301d06 03551d0e 04160414 ed4419c0
d3f0068b eea47bbe 42e72654 c88e3676 300d0609 2a864886 f70d0101 0b050003
82020100 0dae9032 f6a64b7c 44761961 1e2728cd 5e54ef25 bce30890 f929d7ae
6808e194 0058ef2e 2e7e5352 8cb65c07 ea88ba99 8b5094d7 8280df61 090093ad
0d14e6ce c1f23794 78b05f9c b3a273b8 8f059338 cd8d3eb0 b8fbc0cf b1f2ec2d
2d1bccec aa9ab3aa 60821b2d 3bc3843d 578a961e 9c75b8d3 30cd6008 8390d38e
54f14d66 c05d7403 40a3ee85 7ec21f77 9c06e8c1 a7185d52 95edc9dd 259e6dfa
a9eda33a 34d0597b daed50f3 35bfedeb 144d31c7 60f4daf1 879ce248 e2c6c537
fb0610fa 75596631 4729da76 9a1ce982 aeef9ab9 51f78823 9a699562 3ce55580
36d75402 fff1b95d ced4236f d845844a 5b65ef89 0cdd14a7 20cb18a5 25b40df9
01f0a2d2 f400c874 8ea12a48 8e65db13 c4e22517 7debbe87 5b172054 51934a53
030bec5d ca33ed62 fd45c72f 5bdc58a0 8039e6fa d7fe1314 a6ed3d94 4a4274d4
c3775973 cd8f46be 5538effa e89132ea 97580422 de38c3cc bc6dc933 3a6a0a69

```
    3fa0c8ea 728f8c63 8623bd6d 3c969e95 e0494caa a2b92a1b 9c368178 edc3e846
    e2265944 751ed975 8951cd10 849d6160 cb5df997 224d8e98 e6e37ff6 5bbbaecd
    ca4a816b 5e0bf351 e1742be9 7e27a7d9 99494ef8 a580db25 0f1c6362 8ac93367
    6b3c1083 c6addea8 cd168e8d f0073771 9ff2abfc 41f5c18b ec00375d 09e54e80
    effab15c 3806a51b 4ae1dc38 2d3cdcab 1f901ad5 4a9ceed1 706cccee f457f818
    ba846e87
  quit
crypto ca certificate chain _SmartCallHome_ServerCA2
 certificate ca 0509
    308205b7 3082039f a0030201 02020205 09300d06 092a8648 86f70d01 01050500
    3045310b 30090603 55040613 02424d31 19301706 0355040a 13105175 6f566164
    6973204c 696d6974 6564311b 30190603 55040313 1251756f 56616469 7320526f
    6f742043 41203230 1e170d30 36313132 34313832 3730305a 170d3331 31313234
    31383233 33335a30 45310b30 09060355 04061302 424d3119 30170603 55040a13
    1051756f 56616469 73204c69 6d697465 64311b30 19060355 04031312 51756f56
    61646973 20526f6f 74204341 20323082 0222300d 06092a86 4886f70d 01010105
    00038202 0f003082 020a0282 0201009a 18ca4b94 0d002daf 03298af0 0f81c8ae
    4c19851d 089fab29 4485f32f 81ad321e 9046bfa3 86261a1e fe7e1c18 3a5c9c60
    172a3a74 8333307d 615411cb edabe0e6 d2a27ef5 6b6f18b7 0a0b2dfd e93eef0a
    c6b310e9 dcc24617 f85dfda4 daff9e49 5a9ce633 e62496f7 3fba5b2b 1c7a35c2
    d667feab 66508b6d 28602bef d760c3c7 93bc8d36 91f37ff8 db1113c4 9c7776c1
    aeb7026a 817aa945 83e205e6 b956c194 378f4871 6322ec17 6507958a 4bdf8fc6
    5a0ae5b0 e35f5e6b 11ab0cf9 85eb44e9 f80473f2 e9fe5c98 8cf573af 6bb47ecd
    d45c022b 4c39e1b2 95952d42 87d7d5b3 9043b76c 13f1dedd f6c4f889 3fd175f5
    92c391d5 8a88d090 ecdc6dde 89c26571 968b0d03 fd9cbf5b 16ac92db eafe797c
    adebaff7 16cbdbcd 252be51f fb9a9fe2 51cc3a53 0c48e60e bdc9b476 0652e611
    13857263 0304e004 362b2019 02e874a7 1fb6c956 66f07525 dc67c10e 616088b3
    3ed1a8fc a3da1db0 d1b12354 df44766d ed41d8c1 b222b653 1cdf351d dca1772a
    31e42df5 e5e5dbc8 e0ffe580 d70b63a0 ff33a10f ba2c1515 ea97b3d2 a2b5bef2
    8c961e1a 8f1d6ca4 6137b986 7333d797 969e237d 82a44c81 e2a1d1ba 675f9507
```

```
    a32711ee 16107bbc 454a4cb2 04d2abef d5fd0c51 ce506a08 31f991da 0c8f645c
    03c33a8b 203f6e8d 673d3ad6 fe7d5b88 c95efbcc 61dc8b33 77d34432 35096204
    921610d8 9e2747fb 3b21e3f8 eb1d5b02 03010001 a381b030 81ad300f 0603551d
    130101ff 04053003 0101ff30 0b060355 1d0f0404 03020106 301d0603 551d0e04
    1604141a 8462bc48 4c332504 d4eed0f6 03c41946 d1946b30 6e060355 1d230467
    30658014 1a8462bc 484c3325 04d4eed0 f603c419 46d1946b a149a447 3045310b
    30090603 55040613 02424d31 19301706 0355040a 13105175 6f566164 6973204c
    696d6974 6564311b 30190603 55040313 1251756f 56616469 7320526f 6f742043
    41203282 02050930 0d06092a 864886f7 0d010105 05000382 0201003e 0a164d9f
    065ba8ae 715d2f05 2f67e613 4583c436 f6f3c026 0c0db547 645df8b4 72c946a5
    03182755 89787d76 ea963480 1720dce7 83f88dfc 07b8da5f 4d2e67b2 84fdd944
    fc775081 e67cb4c9 0d0b7253 f8760707 4147960c fbe08226 93558cfe 221f6065
    7c5fe726 b3f73290 9850d437 7155f692 2178f795 79faf82d 26876656 3077a637
    78335210 58ae3f61 8ef26ab1 ef187e4a 5963ca8d a256d5a7 2fbc561f cf39c1e2
    fb0aa815 2c7d4d7a 63c66c97 443cd26f c34a170a f890d257 a21951a5 2d9741da
    074fa950 da908d94 46e13ef0 94fd1000 38f53be8 40e1b46e 561a20cc 6f588ded
    2e458fd6 e9933fe7 b12cdf3a d6228cdc 84bb226f d0f8e4c6 39e90488 3cc3baeb
    557a6d80 9924f56c 01fbf897 b0945beb fdd26ff1 77680d35 6423acb8 55a103d1
    4d4219dc f8755956 a3f9a849 79f8af0e b911a07c b76aed34 d0b62662 381a870c
    f8e8fd2e d3907f07 912a1dd6 7e5c8583 99b03808 3fe95ef9 3507e4c9 626e577f
    a75095f7 bac89be6 8ea201c5 d666bf79 61f33c1c e1b9825c 5da0c3e9 d848bd19
    a2111419 6eb2861b 683e4837 1a88b75d 965e9cc7 ef276208 e291195c d2f121dd
    ba174282 97718153 31a99ff6 7d62bf72 e1a3931d cc8a265a 0938d0ce d70d8016
    b478a53a 874c8d8a a5d54697 f22c10b9 bc5422c0 01506943 9ef4b2ef 6df8ecda
    f1e3b1ef df918f54 2a0b25c1 2619c452 100565d5 8210eac2 31cd2e
  quit
crypto ikev1 enable outside
crypto ikev1 policy 10
 authentication pre-share
 encryption aes
```

```
 hash sha
 group 14
 lifetime 86400
telnet timeout 10
ssh stricthostkeycheck
ssh timeout 5
ssh key-exchange group dh-group14-sha256
console timeout 0
console serial
management-access inside
threat-detection basic-threat
threat-detection statistics access-list
no threat-detection statistics tcp-intercept
dynamic-access-policy-record DfltAccessPolicy
tunnel-group 192.168.100.254 type ipsec-l2l
tunnel-group 192.168.100.254 ipsec-attributes
 ikev1 pre-shared-key *****
!
class-map inspection_default
 match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
 parameters
  message-length maximum client auto
  message-length maximum 512
  no tcp-inspection
policy-map global_policy
 class inspection_default
  inspect ip-options
```

```
  inspect netbios
  inspect rtsp
  inspect sunrpc
  inspect tftp
  inspect dns preset_dns_map
  inspect ftp
  inspect h323 h225
  inspect h323 ras
  inspect rsh
  inspect esmtp
  inspect sqlnet
  inspect sip
  inspect skinny
  inspect icmp
  inspect icmp error
policy-map type inspect dns migrated_dns_map_2
 parameters
  message-length maximum client auto
  message-length maximum 512
  no tcp-inspection
policy-map type inspect dns migrated_dns_map_1
 parameters
  message-length maximum client auto
  message-length maximum 512
  no tcp-inspection
!
service-policy global_policy global
prompt hostname context
no call-home reporting anonymous
call-home
```

```
 profile License
  destination address http https://tools.cisco.com/its/service/oddce/services/DDCEService
  destination transport-method http
 profile CiscoTAC-1
  no active
  destination address http https://tools.cisco.com/its/service/oddce/services/DDCEService
  destination address email callhome@cisco.com
  destination transport-method http
  subscribe-to-alert-group diagnostic
  subscribe-to-alert-group environment
  subscribe-to-alert-group inventory periodic monthly
  subscribe-to-alert-group configuration periodic monthly
  subscribe-to-alert-group telemetry periodic daily
Cryptochecksum:b21b9a6a32dbeeccc3423dad14836ec3
: end
```

BR1-CORE - Representative VLAN Gateway and ACL Logic

BR1-CORE#show running-config

Building configuration...

Current configuration : 5460 bytes

!

! Last configuration change at 09:37:31 UTC Tue Jun 9 2026

!

version 17.16

service timestamps debug datetime msec

service timestamps log datetime msec

!

hostname BR1-CORE

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

!

!

!

!

!

!

!

!

!

!

ip dhcp excluded-address 10.2.50.1 10.2.50.10

ip dhcp excluded-address 10.2.60.1 10.2.60.10

```
ip dhcp excluded-address 10.2.99.1 10.2.99.10
ip dhcp excluded-address 10.2.240.1 10.2.240.10
ip dhcp excluded-address 10.2.30.1 10.2.30.10
ip dhcp excluded-address 10.2.31.1 10.2.31.10
ip dhcp excluded-address 10.2.32.1 10.2.32.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 172.16.66.1 172.16.66.10
!
ip dhcp pool BR1_ATM
 network 10.2.50.0 255.255.255.0
 default-router 10.2.50.1
 dns-server 8.8.8.8
!
ip dhcp pool BR1_TELLER
 network 10.2.60.0 255.255.255.0
 default-router 10.2.60.1
 dns-server 8.8.8.8
!
ip dhcp pool BR1_CORP
 network 10.2.240.0 255.255.255.0
 default-router 10.2.240.1
 dns-server 8.8.8.8
!
ip dhcp pool BR1_VOICE
 network 10.2.30.0 255.255.255.0
 default-router 10.2.30.1
!
ip dhcp pool BR1_GUEST
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
```

```
 dns-server 8.8.4.4
!
!
!
ip audit notify log
ip audit po max-events 100
ip cef
login on-success log
no ipv6 cef
!
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-67109008
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-67109008
 revocation-check none
 rsakeypair TP-self-signed-67109008
 hash sha512
!
!
crypto pki certificate chain TP-self-signed-67109008
 certificate self-signed 01
  3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030
  2F312D30 2B060355 04030C24 494F532D 53656C66 2D536967 6E65642D 43657274
  69666963 6174652D 36373130 39303038 301E170D 32363034 32373136 33323037
```

5A170D33 36303432 36313633 3230375A 302F312D 302B0603 5504030C 24494F53

2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D363731 30393030

38308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201 0A028201

0100E427 137B4248 55E4C12E 1BD020CB 6ED7F67E 408E7A03 37C45168
CB66DD79

1030CA21 0E520300 285CBE40 5D44DB0E 804D980A F9BADD00 97189740
52FEC723

3AD90DB0 15EB3B4D A8659FBC 5ACD0B0E FA08BFF5 C27EEA55 9BBBFE49
8F56EC6D

7AE7230A BD04F4B1 98FFE0BF A06AF383 6D592645 C5F03535 0F341034
2AFC93C9

5021A222 15E1B279 6193B544 BDB5C21B ECCAB18D C5DF1A8D 66293109
06A06A34

7942686B 6B5D75BA A501BFED 9B29C257 014B1E62 D7176551 A1B2592E
3E01C65C

4A0F5096 83094171 CC78B2E9 3235D149 7F60427A D09A5183 731EE600 4F2054EA

010462A0 32E9C314 13408E93 44ACF033 0D474E89 C0977E7B 6E28429B
D47F89A2

4DD90203 010001A3 53305130 1D060355 1D0E0416 04147590 A0409BFE 41A4524A

598B3C74 B461844A CDFC301F 0603551D 23041830 16801475 90A0409B FE41A452

4A598B3C 74B46184 4ACDFC30 0F060355 1D130101 FF040530 030101FF 300D0609

2A864886 F70D0101 0D050003 82010100 A0C8A8BF 068D6929 6287BCE9 084A6550

DD3A00EF FF9CB158 EEA1777A 8F008003 9042CF73 14F7B998 5EB42857
AEC06CE5

7709AC7C 11B44EF2 857E1A62 AF286EA7 BA0F4342 C75E1D47 2A6B68DB
E0F8055E

68025FFC 71D262E2 0543E96E 988250BA F174FDCE 6DF3BC99 D823CC3D
19F64CFB

52D4FB51 FF72CDC7 0A518C57 FE29CA53 FE91BF80 9B1C68CD D10C1353
1131439A

BC068EF6 D9DABB8C 19A1DA7B D2F1F781 2E472419 7905C5B3 6B1530DC
A764729F

```
  4DD6F951 09347C9A EACC0FB6 7A46BFD3 A11A7E1A 98ECA58D 954145FA
F7EBB33A
  1B78F9EC 9FBE049D ED08BC9C 9BBD2C03 64D75347 CAD67464 E0FEA1B2
6D7C6C33
  93F0505E B0477813 D9DE154F 03FC0315
        quit
!
!
memory free low-watermark processor 79497
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
vlan internal allocation policy ascending
!
!
!
!
!
interface Ethernet0/0
 description TRUNK_TO_BR1_ACC
 switchport access vlan 60
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 50,60,99,240,300,310,320,400,666
 switchport mode trunk
 spanning-tree portfast trunk
!
interface Ethernet0/1
```

```
 no switchport
 ip address 10.2.0.2 255.255.255.252
!
interface Ethernet0/2
!
interface Ethernet0/3
!
interface Vlan50
 ip address 10.2.50.1 255.255.255.0
 ip access-group ACL_BR_ATM_IN in
!
interface Vlan60
 ip address 10.2.60.1 255.255.255.0
!
interface Vlan99
 ip address 10.2.99.1 255.255.255.0
!
interface Vlan240
 ip address 10.2.240.1 255.255.255.0
!
interface Vlan300
 ip address 10.2.30.1 255.255.255.0
!
interface Vlan310
 ip address 10.2.31.1 255.255.255.0
!
interface Vlan320
 ip address 10.2.32.1 255.255.255.0
!
interface Vlan400
```

```
 ip address 192.168.20.1 255.255.255.0
 ip access-group ACL_BR_GUEST_IN in
!
interface Vlan666
 ip address 172.16.66.1 255.255.255.0
!
ip forward-protocol nd
ip forward-protocol udp
!
!
ip http server
ip http secure-server
ip ssh bulk-mode 131072
ip route 0.0.0.0 0.0.0.0 10.2.0.1
!
ip access-list extended ACL_BR_ATM_IN
 10 permit udp any any eq bootps
 20 permit udp any any eq bootpc
 30 permit icmp 10.2.50.0 0.0.0.255 host 10.2.50.1
 35 permit udp 10.2.50.0 0.0.0.255 host 10.2.99.50 eq syslog
 40 permit ip 10.2.50.0 0.0.0.255 10.1.110.0 0.0.0.255 log
 50 deny ip any any log
ip access-list extended ACL_BR_GUEST_IN
 10 permit udp any any eq domain
 20 permit udp any any eq bootps
 30 deny ip any 10.2.0.0 0.0.255.255 log
 40 deny ip any 10.1.0.0 0.0.255.255 log
 50 permit ip any any
no logging btrace
logging source-interface Ethernet0/1
```

```
logging host 10.2.99.50
!
!
!
control-plane
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input ssh
!
!
end
```

BR1-ACC-USERS:

```
BR1-ACC#show running-config
Building configuration...
Current configuration : 3803 bytes
!
! Last configuration change at 09:58:40 UTC Tue Jun 9 2026
!
version 17.16
service timestamps debug datetime msec
service timestamps log datetime msec
!
```

```
hostname BR1-ACC
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
!
!
!
no ip routing
!
!
!
!
!
!
!
!
!
!
ip audit notify log
ip audit po max-events 100
no ip cef
login on-success log
no ipv6 cef
!
!
!
!
```

!

!

!

!

crypto pki trustpoint TP-self-signed-67109072

 enrollment selfsigned

 subject-name cn=IOS-Self-Signed-Certificate-67109072

 revocation-check none

 rsakeypair TP-self-signed-67109072

 hash sha512

!

!

crypto pki certificate chain TP-self-signed-67109072

 certificate self-signed 01

  3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030

  2F312D30 2B060355 04030C24 494F532D 53656C66 2D536967 6E65642D 43657274

  69666963 6174652D 36373130 39303732 301E170D 32363034 32373137 31353332

  5A170D33 36303432 36313731 3533325A 302F312D 302B0603 5504030C 24494F53

  2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D363731 30393037

  32308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201 0A028201

  0100E47C 37FEFBCD B4B3F0F1 898B59E8 3C6D7C6C 7CA81C0A DF6D9185
B9DBA873

  BFAE4BDD F746FBC8 F44C89BA AF68C00F 338C2963 59B93C93 2CB09205
DBA7CE0F

  1E37F70A FDC1B0F6 89D3F0CC 53D54515 F40B1589 33CAA889 4285B639
D0E99313

  C510550E 800F2060 6B3F07F8 6FE96A65 E082BE0F BB3A58A5 92EE49D8
998E6A3D

  56E6BC7E A873FC84 6EEA211F CB2493AE 914A28E4 301C4997 137D6C36
E7F2BBAC

```
  73D4E427 AEE2D11C A50A9049 CF6A0154 36EE04EF E7228CA7 BC9C01E9
DA513E18
  151782A7 536DD107 AD70AE3B 0FE10D4C 8E9AE107 C1259882 E65D1496
7D9917A8
  65988429 B7AB426B 96469C2D 61137BFD 949ACE4F CD950C33 299A61A7
32ECD5DD
  274B0203 010001A3 53305130 1D060355 1D0E0416 0414CC82 56BDB427 14812FD7
  CF01BAB8 C3268E5B FF19301F 0603551D 23041830 168014CC 8256BDB4 2714812F
  D7CF01BA B8C3268E 5BFF1930 0F060355 1D130101 FF040530 030101FF 300D0609
  2A864886 F70D0101 0D050003 82010100 8B5DE8F9 33FB99CD DF71A853
64EC4428
  FBB9984C E223461E A16FDE62 2264B9A5 2057B2EB B8A4BE49 810E620E
8092F91D
  0AF5C192 F07428FD D7FCF952 FF60C5EB 171D2FEF 35B64333 6489D1AD
1AFED074
  A4D09AB0 BCC18DE4 865696C3 4C3D851C 82EEB232 F9A74670 C7E701AB
0EBE068C
  4072F5D4 D1354389 46B7F4C1 E7B94F8B 96E6C450 0F2446B2 4DB0274E
F152B20D
  739FF2BD 59DEA5B3 F20A527C BF3592CD 54D7FB47 53BFDC1D 4C67FA39
A89DBDE8
  CF8F46F9 51D65608 8838BB20 DF35299A E519A686 45DDCA89 99753E91
7A94EDDA
  13EC7E6B 7A837D6E 10F1387F F2396A85 73926B48 CE768D3B 3AADC391
BD1E3CAB
  32B69B18 6819F69B C08DFDD2 1FEAEB58
      quit
!
!
memory free low-watermark processor 79497
!
!
spanning-tree mode rapid-pvst
```

```
spanning-tree extend system-id
!
!
vlan internal allocation policy ascending
!
!
!
!
!
interface Ethernet0/0
 description UPLINK_TO_BR1_CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/1
 switchport access vlan 60
 switchport mode access
 spanning-tree portfast
!
interface Ethernet0/2
 switchport access vlan 50
 switchport mode access
 spanning-tree portfast
!
interface Ethernet0/3
 description GUEST_AP_BAGLANTISI
 switchport access vlan 400
 switchport mode access
 spanning-tree portfast
!
```

```
interface Ethernet1/0
 description BR1_SYSLOG_SERVER
 switchport access vlan 99
 switchport mode access
 spanning-tree portfast
!
interface Ethernet1/1
!
interface Ethernet1/2
!
interface Ethernet1/3
!
interface Vlan99
 ip address 10.2.99.2 255.255.255.0
 no ip route-cache
!
ip default-gateway 10.2.99.1
ip forward-protocol nd
ip forward-protocol udp
!
!
ip http server
ip http secure-server
ip ssh bulk-mode 131072
!
no logging btrace
logging source-interface Vlan99
logging host 10.2.99.50
!
!
```

!

control-plane

!

!

!

line con 0

 logging synchronous

line aux 0

line vty 0 4

 login

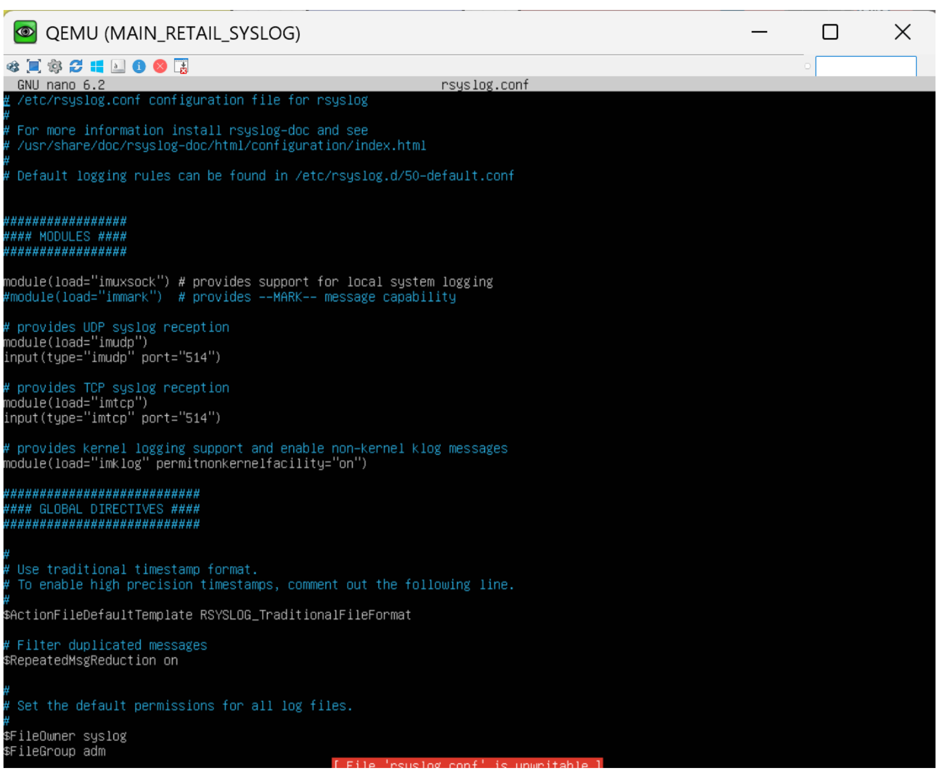
 transport input ssh

!

!

End

MAIN-RETAIL-SYSLOG:



GUEST-AP:

GUEST-AP#show running-config

Building configuration...

Current configuration : 3266 bytes

!

! Last configuration change at 09:59:34 UTC Tue Jun 9 2026

!

version 17.16

service timestamps debug datetime msec

service timestamps log datetime msec

!

hostname GUEST-AP

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

!

!

!

!

!

!

!

!

!

!

```
!
!
!
ip audit notify log
ip audit po max-events 100
ip cef
login on-success log
no ipv6 cef
!
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-67109120
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-67109120
 revocation-check none
 rsakeypair TP-self-signed-67109120
 hash sha512
!
!
crypto pki certificate chain TP-self-signed-67109120
 certificate self-signed 01
  3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030
  2F312D30 2B060355 04030C24 494F532D 53656C66 2D536967 6E65642D 43657274
  69666963 6174652D 36373130 39313230 301E170D 32363036 30393039 35393534
  5A170D33 36303630 38303935 3935345A 302F312D 302B0603 5504030C 24494F53
```

2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D363731 30393132

30308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201 0A028201

0100BB8C 82979B65 42B8CEEE 32D173EF 5B93AE15 694B92DD 871B7310
A8ED0A4F

4AEEA817 5B8E0342 F0CCF6D6 4582063E 7A2F6566 A7F1135E 95866A3D
A1E8CA58

E36F8376 45963D74 5B164E68 7C2BCFE4 ECD131A9 BB1C16F4 7368F646
3F6A2656

73290C99 CEDA4339 F703EC88 9F08FFB9 E4E78BA8 D9F08971 DE0969A6
CFC04037

50CC0C1D 4052652B CF008723 8784D1BC F2CE5E01 E998F5FC CA757BC5
58F7D928

EAE889A4 40DC2D8F 995EC236 94843BE0 3C312451 A2B45923 390A2CC3
565C769B

5A037765 B3E400BA C5C20456 4CFB90A8 61FB774E 85B6A161 45ECF818
F86FD6D7

582DB01E BF928BC6 FEC6580C B584F661 569CA2B3 B8B4AD3A 1EBAA874
50123248

6ADB0203 010001A3 53305130 1D060355 1D0E0416 0414EAB8 C1356A87
FF6E330E

A0BADB1C 11FE7D34 15BB301F 0603551D 23041830 168014EA B8C1356A
87FF6E33

0EA0BADB 1C11FE7D 3415BB30 0F060355 1D130101 FF040530 030101FF
300D0609

2A864886 F70D0101 0D050003 82010100 345CAB15 2BB650D7 A00ED94B
394C6821

9D0BF8AE B8747BD6 454E9D83 62F0D0F1 494AB469 3526BA27 BFBED803
C01135A5

9265F6F8 FB65C056 E137F299 D8F43AD4 4C4CD401 1CF4C546 12F0D3E9
1D33BF2C

D2BFACBD 19C92013 B6A03216 FEA85D76 7CCC9964 E0CAECEF 7EF55C11
57F8C927

D2988C43 80BA84E9 8B66506F 4F7BD274 D809E473 28C5AD39 4F9E78DC
40054D9D

```
  EDA2BAE1 416E46BA B9FE1CA4 A81A3F67 155CBA79 A5086A73 0274D81F
7BDF1093
  F536171E C860ED80 415CE31C 891CC9F2 FF18F5B0 361E86B8 681C59D0
4FAA7653
  7970646E 7E5CCC71 8BAF289D DEB1B5A7 C486A618 3E304D1D 1542B33E
5041FF7B
  AD9173D1 D7C6FC6F 86203F75 05864EE9
      quit
!
!
memory free low-watermark processor 79497
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
vlan internal allocation policy ascending
!
!
!
!
!
interface Ethernet0/0
 description UPLINK_TO_ACC-USERS
 switchport access vlan 400
 switchport mode access
!
interface Ethernet0/1
 description MOBILE_PHONE_WIFI
```

```
 switchport access vlan 400
 switchport mode access
 spanning-tree portfast
!
interface Ethernet0/2
!
interface Ethernet0/3
!
ip forward-protocol nd
ip forward-protocol udp
!
!
ip http server
ip http secure-server
ip ssh bulk-mode 131072
!
no logging btrace
!
!
!
control-plane
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input ssh
```

!

!

end

BR2-RTR - Satellite Branch 1 Router Summary

BR2-RTR#sh running-config

*Jun 9 10:12:48.132: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.100.254 on Tunnel1 from LOADING to FULL, Loading Done

BR2-RTR#sh running-config

Building configuration...

Current configuration : 5364 bytes

!

version 17.16

service timestamps debug datetime msec

service timestamps log datetime msec

!

hostname BR2-RTR

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

!

!

!

!

!

```
!
!
!
!
!
ip dhcp excluded-address 10.3.50.1 10.3.50.10
ip dhcp excluded-address 10.3.60.1 10.3.60.10
ip dhcp excluded-address 10.3.30.1 10.3.30.10
ip dhcp excluded-address 10.3.32.1 10.3.32.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
!
ip dhcp pool POOL_BR2_ATM
 network 10.3.50.0 255.255.255.0
 default-router 10.3.50.1
 dns-server 10.1.110.10 8.8.8.8
 domain-name securetrust.local
!
ip dhcp pool POOL_BR2_TELLER
 network 10.3.60.0 255.255.255.0
 default-router 10.3.60.1
 dns-server 10.1.110.10 8.8.8.8
 domain-name securetrust.local
!
ip dhcp pool POOL_BR2_VOICE
 network 10.3.30.0 255.255.255.0
 default-router 10.3.30.1
 option 150 ip 10.1.100.20
 dns-server 10.1.110.10
!
ip dhcp pool POOL_BR2_CCTV
```

```
 network 10.3.32.0 255.255.255.0
 default-router 10.3.32.1
 dns-server 10.1.110.10
!
ip dhcp pool POOL_BR2_GUEST
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8 8.8.4.4
!
!
!
no ip cef
login on-success log
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-67109200
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-67109200
 revocation-check none
 rsakeypair TP-self-signed-67109200
 hash sha512
```

!

!

crypto pki certificate chain TP-self-signed-67109200

certificate self-signed 01

3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030

2F312D30 2B060355 04030C24 494F532D 53656C66 2D536967 6E65642D 43657274

69666963 6174652D 36373130 39323030 301E170D 32363035 31323131 31313034

5A170D33 36303531 31313131 3130345A 302F312D 302B0603 5504030C 24494F53

2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D363731 30393230

30308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201 0A028201

0100A7D8 8DEE65DD ADE5449F 2E2A98AE 879AF6A9 732C6451 9106B49C
71F8B1DF

0EAF8E3E 6EECEEE4 BE50C6C5 5E4C4382 34CC7628 FCE10C40 B6E492E4
78F97F65

DA026A09 CD092C78 C5C096CF 75656EA3 F3E94F69 4F774E78 4D065970
CF4BF9C8

F4A6480E 40AA6305 D5C2E78E B404B9D1 1B0AEF0A 463E8D07 70220412
8B59E1C4

9AC799BF 225C53DA 03B2CB2C 91E6DF75 623796E2 A3A1C16F B659F0FF
73592C0E

66EF60CE 6EBA5BE1 21D22A7D DA325A53 70AF5806 0591B404 42B2014B
DC402464

6A75BE17 7EBDD588 36DF7BD0 619F0C46 E15C135F 258621A1 F72B01B3
FCC56415

EA150F14 5681AA3A 1BDDFF53 D1FA60E2 DC21663B F097C077 9A75A638
1317C6F0

4A990203 010001A3 53305130 1D060355 1D0E0416 0414EA35 F24B268F 7515C58B

24F5E9A5 97DF2A89 8D26301F 0603551D 23041830 168014EA 35F24B26 8F7515C5

8B24F5E9 A597DF2A 898D2630 0F060355 1D130101 FF040530 030101FF 300D0609

2A864886 F70D0101 0D050003 82010100 4C34EE09 6110C8C8 70E5A31F
C1E9AEA4

```
 601FDCF7 09CF2E6E F49B005D 19554F1A 9D011394 1036F768 56F3748B 6AB09AB3
 7E79C1C0 67F7B0BD E918D8F2 701ECD94 3B91CA2B 68B62434 C71A4AAE D6AD9006
 4D22EB45 BD7A8DE6 81002606 8F39F2A7 3E6C30EB 2EBFB675 D25B1F9E F1A75B64
 0DAA57CD 0CFD841B DC314DF1 7F522F16 E9E1FD01 26C958F8 92A7B6B4 194B41A5
 E809D75E 81E53F82 4ACF454B 2302B09C 6DEE583D 4CA279D2 C65BD639 B61F7685
 55753AA6 7039E184 7D2D1638 0D4A2487 AB214129 76E50E36 0EFA27AF 7A2F30DD
 1109E53E 86BCF3D7 AD668851 45A53CCB EEAA48C6 5CE2B971 61F2927C 8C38915F
 77EC276D 1033332C BC20838D 76DF514F
    quit
!
!
memory free low-watermark processor 79983
!
!
spanning-tree mode rapid-pvst
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
!
crypto isakmp policy 10
 encryption aes
 hash sha
 authentication pre-share
 group 14
crypto isakmp key fkgbankk address 192.168.100.254
!
!
crypto ipsec transform-set TS esp-aes esp-sha-hmac
 mode tunnel
!
crypto ipsec profile IPSEC_VTI_PROFILE
 set transform-set TS
!
!
!
!
!
!
!
!
interface Tunnel1
 description VTI_TO_HQ
 ip address 10.255.2.2 255.255.255.252
 ip ospf mtu-ignore
```

```
 tunnel source Ethernet0/0
 tunnel mode ipsec ipv4
 tunnel destination 192.168.100.254
 tunnel protection ipsec profile IPSEC_VTI_PROFILE
!
interface Ethernet0/0
 ip address 192.168.100.252 255.255.255.0
!
interface Ethernet0/1
 description TRUNK_TO_BR2_SW
 no ip address
!
interface Ethernet0/1.60
 encapsulation dot1Q 60
 ip address 10.3.60.1 255.255.255.0
!
interface Ethernet0/1.99
 description Management
 encapsulation dot1Q 99
 ip address 10.3.99.1 255.255.255.0
!
interface Ethernet0/1.300
 description Voice
 encapsulation dot1Q 300
 ip address 10.3.30.1 255.255.255.0
!
interface Ethernet0/1.400
 description Guest_WiFi
 encapsulation dot1Q 400
 ip address 192.168.30.1 255.255.255.0
```

```
!
interface Ethernet0/2
 description TRUNK_TO_NEW_SWITCH
 no ip address
!
interface Ethernet0/2.50
 description ATM_KIOSK_GATEWAY
 encapsulation dot1Q 50
 ip address 10.3.50.1 255.255.255.0
!
interface Ethernet0/2.320
 description CCTV_GATEWAY
 encapsulation dot1Q 320
 ip address 10.3.32.1 255.255.255.0
!
interface Ethernet0/3
 no ip address
 shutdown
!
router ospf 1
 router-id 10.3.99.1
 network 10.3.0.0 0.0.255.255 area 0
 network 10.255.2.0 0.0.0.3 area 0
 network 192.168.30.0 0.0.0.255 area 0
!
ip forward-protocol nd
ip forward-protocol udp
!
!
ip http server
```

```
ip http secure-server
ip route 0.0.0.0 0.0.0.0 Tunnel1
ip ssh bulk-mode 131072
no logging btrace
!
!
!
control-plane
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input ssh
!
!
!
!
end
```

BR3-RTR - Satellite Branch 2 Router Summary

```
BR3-RTR#show running-config
Building configuration...
Current configuration : 5592 bytes
!
version 17.16
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
!
hostname BR3-RTR
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
!
!
!
!
!
!
!
!
!
 --More--
*Jun  9 10:08:47.746: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.100.254 on Tunnel1
from LOADING to FULL, Loading Done
!
ip dhcp excluded-address 10.4.50.1 10.4.50.10
ip dhcp excluded-address 10.4.60.1 10.4.60.10
ip dhcp excluded-address 10.4.30.1 10.4.30.10
ip dhcp excluded-address 10.4.32.1 10.4.32.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
!
ip dhcp pool POOL_BR2_ATM
 network 10.4.50.0 255.255.255.0
```

```
 default-router 10.4.50.1
 dns-server 10.1.110.10 8.8.8.8
 domain-name securetrust.local
!
ip dhcp pool POOL_BR2_TELLER
 network 10.4.60.0 255.255.255.0
 default-router 10.4.60.1
 dns-server 10.1.110.10 8.8.8.8
 domain-name securetrust.local
!
ip dhcp pool POOL_BR2_VOICE
 network 10.4.30.0 255.255.255.0
 default-router 10.4.30.1
 option 150 ip 10.1.100.20
 dns-server 10.1.110.10
!
ip dhcp pool POOL_BR2_CCTV
 network 10.4.32.0 255.255.255.0
 default-router 10.4.32.1
 dns-server 10.1.110.10
!
ip dhcp pool POOL_BR2_GUEST
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8 8.8.4.4
!
!
!
no ip cef
login on-success log
```

```
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-67109280
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-67109280
 revocation-check none
 rsakeypair TP-self-signed-67109280
 hash sha512
!
!
crypto pki certificate chain TP-self-signed-67109280
 certificate self-signed 01
  3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030
  2F312D30 2B060355 04030C24 494F532D 53656C66 2D536967 6E65642D 43657274
  69666963 6174652D 36373130 39323830 301E170D 32363035 31343133 32313330
  5A170D33 36303531 33313332 3133305A 302F312D 302B0603 5504030C 24494F53
  2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D363731 30393238
  30308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201 0A028201
  0100B875 B7AEF261 F139E788 F24F7FDD B3336FCF E2A69A8C 86F53E43
906D4405
```

6ED7AD9F 9874C035 975E3CF5 403F2B1D E5A0F7CD 039BC83D 879D6792 D39A4A17

50DF4786 741F2801 5FEF1C28 BAA0F4D7 16DF15C9 664C0CCE 0BD7E75C B6AE4371

CBE49D4E 33ECA152 56B98B43 A5E3CF80 D8BE0D5F 878D3145 1FA63284 01F98B1F

338D4EEC 0D5A1C01 C3986C8E D3CAB925 D1938C1B 986C9979 45508EDC 585C3C38

1994BDC4 EC3353C5 B418C249 57002CC8 E103505F 7427AC26 A7ECC1C1 CEB55F0A

46A52610 04530F98 D1FBB420 8A588818 9037473F 3FAB50CA 75D93123 3237EFB2

64F5D050 141DE4B5 7EB6235F DD095587 F61DEF23 AD140250 6D53A6FC F027082F

950B0203 010001A3 53305130 1D060355 1D0E0416 0414B681 B42CAE71 68D7BC79

CDAD4DB3 8C80AA7E F86A301F 0603551D 23041830 168014B6 81B42CAE 7168D7BC

79CDAD4D B38C80AA 7EF86A30 0F060355 1D130101 FF040530 030101FF 300D0609

2A864886 F70D0101 0D050003 82010100 4CD940E8 5D0DDF28 0D907CC4 E824166B

53B964E3 C0044275 D57FB0BD 1BA2AEDD B22C43AF 13BEC5F2 8880064C E878001D

654F3E33 25CA244D CD1F4C7C DEE122E0 A37BE7B2 2AEF6CCB D1E8E7B0 9D9AB8C6

6782AFC9 22355EF4 D668C643 D2527E7D 7B65BFCB E7800D26 A53A9D27 4558F5FF

E4A7AE38 53FB727E 2AA90731 19F2E92B AD17E51D 95C59BF3 5FB15239 BB2F8596

4B99EDA3 F0CA995F 556527C0 A9897895 BD3C2331 1937ACE7 FA97CE41 F25DA90F

C4C63DB1 C2592961 AA0D746F 7251A77C 181CA105 FBDA5767 801EAE67 620A7C2B

1612C958 ED02DCD4 AC22C15F 807A02C4 63B5CF07 415DE3F8 13313A08 F3861AAE

```
  78911859 2FE890DE 651ADABE 83A9ACA6
        quit
!
!
memory free low-watermark processor 79983
!
!
spanning-tree mode rapid-pvst
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
crypto isakmp policy 10
 encryption aes
 hash sha
 authentication pre-share
 group 14
crypto isakmp key fkgbank address 192.168.100.254
!
```

```
!
crypto ipsec transform-set TS esp-aes esp-sha-hmac
 mode tunnel
!
crypto ipsec profile IPSEC_VTI_PROFILE
 set transform-set TS
!
!
!
!
!
!
!
!
interface Tunnel1
 description VTI_TO_HQ
 ip address 10.255.3.2 255.255.255.252
 tunnel source Ethernet0/0
 tunnel mode ipsec ipv4
 tunnel destination 192.168.100.254
 tunnel protection ipsec profile IPSEC_VTI_PROFILE
!
interface Ethernet0/0
 description WAN_TO_HQ_VPN
 ip address 192.168.100.251 255.255.255.0
!
interface Ethernet0/1
 description TRUNK_TO_BR3_SW
 no ip address
!
```

```
interface Ethernet0/1.50
 description ATM_KIOSK_GATEWAY
 encapsulation dot1Q 50
 ip address 10.4.50.1 255.255.255.0
!
interface Ethernet0/1.60
 description TELLER_OPS_GATEWAY
 encapsulation dot1Q 60
 ip address 10.4.60.1 255.255.255.0
!
interface Ethernet0/1.99
 description MGMT_NETWORK_GATEWAY
 encapsulation dot1Q 99
 ip address 10.4.99.1 255.255.255.0
!
interface Ethernet0/1.240
 description CORP_OFFICE_GATEWAY
 encapsulation dot1Q 240
 ip address 10.5.240.1 255.255.255.0
!
interface Ethernet0/1.300
 description VOICE_VOIP_GATEWAY
 encapsulation dot1Q 300
 ip address 10.4.30.1 255.255.255.0
!
interface Ethernet0/2
 description TRUNK_TO_CCTV_PRINTER_SW
 no ip address
!
interface Ethernet0/2.310
```

```
 description PRINTER_GATEWAY
 encapsulation dot1Q 310
 ip address 10.4.31.1 255.255.255.0
!
interface Ethernet0/2.320
 description CCTV_GATEWAY
 encapsulation dot1Q 320
 ip address 10.4.32.1 255.255.255.0
!
interface Ethernet0/3
 no ip address
 shutdown
!
router ospf 1
 router-id 10.4.99.1
 network 10.4.0.0 0.0.255.255 area 0
 network 10.5.0.0 0.0.255.255 area 0
 network 10.255.3.0 0.0.0.3 area 0
 network 192.168.40.0 0.0.0.255 area 0
!
ip forward-protocol nd
ip forward-protocol udp
!
!
ip http server
ip http secure-server
ip route 0.0.0.0 0.0.0.0 Tunnel1
ip ssh bulk-mode 131072
no logging btrace
!
```

```
!
!
control-plane
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input ssh
!
!
!
!
end
```

BR3-ACC-USER:

```
BR3-ACC-USERS#show running-config
Building configuration...

Current configuration : 3926 bytes
!
! Last configuration change at 10:10:03 UTC Tue Jun 9 2026
!
version 17.16
service timestamps debug datetime msec
service timestamps log datetime msec
```

```
!
hostname BR3-ACC-USERS
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
!
!
!
!
!
!
!
!
!
!
!
!
!
ip audit notify log
ip audit po max-events 100
ip cef
login on-success log
no ipv6 cef
!
!
!
!
```

```
!
!
!
!
crypto pki trustpoint TP-self-signed-67109312
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-67109312
 revocation-check none
 rsakeypair TP-self-signed-67109312
 hash sha512
!
!
crypto pki certificate chain TP-self-signed-67109312
 certificate self-signed 01
  3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030
  2F312D30 2B060355 04030C24 494F532D 53656C66 2D536967 6E65642D 43657274
  69666963 6174652D 36373130 39333132 301E170D 32363035 31323039 32373038
  5A170D33 36303531 31303932 3730385A 302F312D 302B0603 5504030C 24494F53
  2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D363731 30393331
  32308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201 0A028201
  0100ACE3 2EE6F456 DE251694 A4F0E717 ED51C763 FE37D527 AEF06C51
C4AAFF15
  E1D6F456 7F0DD4BD 4CB48526 6050879A 7AC88D48 4D1A4F79 A223B6F9
99AD9543
  96AA1449 E0B520CA E175F639 91FC0DD0 4BEA564D 5D2EC0E8 F47D2A49
A0C4A6B8
  2BDB1B1C 346B7EB9 1E7E3962 1A58F831 B480854A 6E1B61FB 2A57956F
C437C088
  5F6AC852 0F43943F CC2BE05F A7EDD0CD 4B408BFB F9C9FE1B EA9A3AFE
EF181854
```

ACDC46C0 46A88003 52BE02B5 0F5C6D93 BE27FFD0 5BB58EC0 006F5E1F C0827CE3
C928C623 98CF45DB 3C17B5D9 7D1946EB 0A208661 74648803 6A963FCC AB772016
BFB7674A C3CB4BAB 9E99DD1B ACC4712F B7870942 1C95BA8E 25B13151 E409DEA0
CBC50203 010001A3 53305130 1D060355 1D0E0416 0414C345 58C9B328 A2491FE9
46B67FDC 86298914 9FCC301F 0603551D 23041830 168014C3 4558C9B3 28A2491F
E946B67F DC862989 149FCC30 0F060355 1D130101 FF040530 030101FF 300D0609
2A864886 F70D0101 0D050003 82010100 21774EE0 9ABE94D4 86779C1A E9A53215
8B625872 3D35B187 EB2C7557 212DEF7E EC23AD84 21D5646A 85B21888 B16FD69B
35F87912 36FEEB47 2D94D691 EAAC0BAE FCC1EC9D 30A0C8CE C1732E0A 23AFF8A4
914DFF7A 5246BEC5 5E9B3CDA 9250ECA6 46E88899 D30A0EB6 236074DE E8D9B8E5
EE5DA014 18F87788 D2DE9C71 66C115F1 E5CAEA74 70EF6FB4 88FF5DAE 2CD11C01
87EAAC49 F975472F 80D489FF 83C13AF9 512D43B0 5125C82B 540B1275 E3F5A1A0
625F405D 26D258D6 B953648B 929BC896 8623B8EE 2974F246 2E48966D 73E1E5AB
AC330BE0 F638318D 5AD59842 88551554 0D0CEEDF 7C3A1E4D AB808EE0 62569333
9A5F7401 68B15716 AB3B141C B13C1D05

quit

!

!

memory free low-watermark processor 79497

!

!

spanning-tree mode rapid-pvst

```
spanning-tree extend system-id
!
!
vlan internal allocation policy ascending
!
!
!
!
!
interface Ethernet0/0
 description TRUNK_TO_RTR
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 50,60,99,240,300
 switchport mode trunk
!
interface Ethernet0/1
 description BR3_ATM
 switchport access vlan 50
 switchport mode access
 spanning-tree portfast
!
interface Ethernet0/2
 description BR3_PC_OFFICE
 switchport access vlan 60
 switchport mode access
 spanning-tree portfast
!
interface Ethernet0/3
 description BR3_PC_TELLER
 switchport access vlan 60
```

```
 switchport mode access
 spanning-tree portfast
!
interface Ethernet1/0
 description BR3_VOIP_PHONE
 switchport access vlan 300
 switchport mode access
 spanning-tree portfast
!
interface Ethernet1/1
 description UNUSED
 shutdown
!
interface Ethernet1/2
 description UNUSED
 shutdown
!
interface Ethernet1/3
 description UNUSED
 shutdown
!
interface Vlan99
 description BR3_aCC_USERS_MANAGEMENT
 ip address 10.4.99.2 255.255.255.0
!
ip default-gateway 10.5.99.1
ip forward-protocol nd
ip forward-protocol udp
!
!
```

ip http server

ip http secure-server

ip ssh bulk-mode 131072

!

no logging btrace

!

!

!

control-plane

!

!

!

line con 0

 logging synchronous

line aux 0

line vty 0 4

 login

 transport input ssh

!

!

end

GUEST_PHONE can only connect to the internet and ping:

```
DORA@@ IP 192.168.20.11/24 GW 192.168.20.1

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=120 time=6.612 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=120 time=2.889 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=120 time=2.709 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=120 time=3.048 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=120 time=2.536 ms

VPCS> ping 10.1.240.11

*192.168.20.1 icmp_seq=1 ttl=255 time=1.233 ms (ICMP type:3, code:13, Communicat
ion administratively prohibited)
*192.168.20.1 icmp_seq=2 ttl=255 time=1.059 ms (ICMP type:3, code:13, Communicat
ion administratively prohibited)
*192.168.20.1 icmp_seq=3 ttl=255 time=1.222 ms (ICMP type:3, code:13, Communicat
ion administratively prohibited)
*192.168.20.1 icmp_seq=4 ttl=255 time=1.275 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.20.1 icmp_seq=5 ttl=255 time=1.188 ms (ICMP type:3, code:13, Communication administratively prohibited)
```

However, Marketing PCs can also ping admins:

```
DDORA IP 10.1.240.12/24 GW 10.1.240.1

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=120 time=2.894 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=120 time=2.374 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=120 time=2.145 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=120 time=2.667 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=120 time=2.852 ms

VPCS> ping 10.1.200.10

84 bytes from 10.1.200.10 icmp_seq=1 ttl=63 time=1.167 ms
84 bytes from 10.1.200.10 icmp_seq=2 ttl=63 time=1.222 ms
84 bytes from 10.1.200.10 icmp_seq=3 ttl=63 time=1.165 ms
84 bytes from 10.1.200.10 icmp_seq=4 ttl=63 time=0.970 ms
84 bytes from 10.1.200.10 icmp_seq=5 ttl=63 time=1.184 ms
```

PC-DMZ-WEB:

```
VPCS> sh ip

NAME        : VPCS[1]
IP/MASK     : 10.1.100.11/24
GATEWAY     : 10.1.100.1
DNS         :
MAC         : 00:50:79:66:68:03
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS>
```

REMOTE-WORKER:

VPN connection:

```
user@user-pc:~/Desktop$ sudo ipsec up fkgbank-vpn
initiating IKE_SA fkgbank-vpn[2] to 192.168.100.254
generating IKE_SA_INIT request 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N(HASH_ALG) N(REDIR_SUP) ]
sending packet: from 192.168.100.50[500] to 192.168.100.254[500] (936 bytes)
received packet: from 192.168.100.254[500] to 192.168.100.50[500] (38 bytes)
parsed IKE_SA_INIT response 0 [ N(INVAL_KE) ]
peer didn't accept DH group ECP_256, it requested MODP_2048
initiating IKE_SA fkgbank-vpn[2] to 192.168.100.254
generating IKE_SA_INIT request 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N(HASH_ALG) N(REDIR_SUP) ]
sending packet: from 192.168.100.50[500] to 192.168.100.254[500] (1128 bytes)
received packet: from 192.168.100.254[500] to 192.168.100.50[500] (619 bytes)
parsed IKE_SA_INIT response 0 [ SA KE No V V N(NATD_S_IP) N(NATD_D_IP) CERTREQ N(FRAG_SUP) V ]
received Cisco Delete Reason vendor ID
received Cisco Copyright (c) 2009 vendor ID
received FRAGMENTATION vendor ID
selected proposal: IKE:AES_CBC_256/HMAC_SHA2_256_128/PRF_HMAC_SHA2_256/MODP_2048
received 2 cert requests for an unknown ca
authentication of 'FKG_VPN' (myself) with pre-shared key
establishing CHILD_SA fkgbank-vpn{3}
generating IKE_AUTH request 1 [ IDi N(INIT_CONTACT) IDr AUTH CPRQ(ADDR DNS) SA TSi TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) N(EAP_ONLY) N(MSG_ID_SYN_SU
) ]
sending packet: from 192.168.100.50[4500] to 192.168.100.254[4500] (336 bytes)
received packet: from 192.168.100.254[4500] to 192.168.100.50[4500] (288 bytes)
parsed IKE_AUTH response 1 [ V IDr AUTH CPRP(ADDR DNS) SA TSi TSr N(ESP_TFC_PAD_N) N(NON_FIRST_FRAG) N(MOBIKE_SUP) ]
authentication of '192.168.100.254' with pre-shared key successful
IKE_SA fkgbank-vpn[2] established between 192.168.100.50[FKG_VPN]...192.168.100.254[192.168.100.254]
scheduling reauthentication in 10056s
maximum IKE_SA lifetime 10596s
adding DNS server failed
adding DNS server failed
handling INTERNAL_IP4_DNS attribute failed
installing new virtual IP 10.1.220.10
received ESP_TFC_PADDING_NOT_SUPPORTED, not using ESPv3 TFC padding
selected proposal: ESP:AES_CBC_256/HMAC_SHA2_256_128/NO_EXT_SEQ
CHILD_SA fkgbank-vpn{3} established with SPIs ceed8d52_i 62391bf6_o and TS 10.1.220.10/32 === 10.0.0.0/8
peer supports MOBIKE
connection 'fkgbank-vpn' established successfully
```

VPN disconnection:

```
user@user-pc:~/Desktop$ sudo ipsec down fkgbank-vpn
deleting IKE_SA fkgbank-vpn[1] between 192.168.100.50[FKG_VPN]...192.168.100.254[192.168.100.254]
sending DELETE for IKE_SA fkgbank-vpn[1]
generating INFORMATIONAL request 3 [ D ]
sending packet: from 192.168.100.50[4500] to 192.168.100.254[4500] (80 bytes)
received packet: from 192.168.100.254[4500] to 192.168.100.50[4500] (80 bytes)
parsed INFORMATIONAL response 3 [ D ]
IKE_SA deleted
IKE_SA [1] closed successfully
```

APPENDIX C - FINAL TEST MATRIX

Test	Command / Method	Final Interpretation
HQ-ASA to BR tunnel IPs	ping 10.255.1.2 / 10.255.2.2 / 10.255.3.2	Successful
BR routers/ASA to HQ core	ping 10.1.0.2 source VLAN interfaces	Successful for authorized VLANs
HQ-CORE to BR1 Teller	ping 10.2.60.x	Successful
HQ-CORE to BR1 ATM	ping 10.2.50.x	Restricted by ATM ACL if policy applies
HQ-CORE to BR2 Teller	ping 10.3.60.11 / 10.3.60.12	Successful after fixes
HQ-CORE to BR2 ATM	ping 10.3.50.11 / 10.3.50.12	Successful or policy-controlled depending on ACL
HQ-CORE to BR3 Teller/ATM/Voice	ping 10.4.60.x / 10.4.50.x / 10.4.30.x	Successful for authorized endpoints
BR1 to BR2/BR3 Teller	endpoint pings through HQ	Successful
BR2/BR3 to BR1 Teller	endpoint pings through HQ	Successful after BR1-ASA ACL update
Guest to internal sites	ping 10.1/10.2/10.3/10.4 internal targets	Blocked by policy
DMZ local connectivity	ASA to PC-DMZ-WEB and WEB-SERVER	Successful after duplicate IP correction

APPENDIX D - SUPPLEMENTAL SOURCE LIST FROM TEZ_KAYNAK

The uploaded TEZ_KAYNAK.zip file included two PDF papers and a LINKLER.txt file. The most directly relevant sources used in this report are the enterprise network architecture paper and the IPsec virtual enterprise network paper. Additional links can be reviewed for broader research context if the final submission requires more literature depth.

[1] https://upcommons.upc.edu/entities/publication/823e186f-7d1d-4e5c-98d4-122f80b800cc

[2] https://dspace.univ-ouargla.dz/jspui/handle/123456789/39932

[3] https://www.milestoneresearch.in/JOURNALS/index.php/IJHCI/article/view/150

[4] https://www.sciencedirect.com/science/article/pii/S0957417423027963

[5] https://www.sciencedirect.com/science/article/abs/pii/S0140366498001583

[6] https://www.researchgate.net/publication/284503626_A_design_science_research_methodology_for_information_systems_research

[7] https://www.jvcbm.ir/article_196269_en.html

[8] https://www.sciencedirect.com/science/article/pii/S0038012125000655

[9] https://journals.co.za/doi/abs/10.7166/35-2-2986

[10] https://www.researchgate.net/profile/M-A-Jobayer-Bin-Bakkre/publication/285537089_Design_and_Simulation_of_a_Banking_Network_System/links/565ee71e08aefe619b27469a/Design-and-Simulation-of-a-Banking-Network-System.pdf

[11] http://lib.buet.ac.bd:8080/xmlui/bitstream/handle/123456789/2869/Full%20%20Thesis%20.pdf?sequence=1

[12] https://www.sciencedirect.com/science/article/abs/pii/S0377221704005168

[13] https://www.sciencedirect.com/science/article/pii/S0038012125000655

[14] https://www.researchgate.net/profile/M-A-Jobayer-Bin-Bakkre/publication/285537089_Design_and_Simulation_of_a_Banking_Network_System/links/565ee71e08aefe619b27469a/Design-and-Simulation-of-a-Banking-Network-System.pdf

APPENDIX E - OPERATIONAL RUNBOOK

This appendix provides a practical runbook for operating and troubleshooting the FKG Bank topology during a live demonstration or final defense. The runbook is intentionally written as a checklist so that the team can quickly verify the health of the network before presenting it to the jury.

E.1 Pre-Demo Health Checklist

10. Start HQ-ASA, HQ-CORE and all branch edge devices first.
11. Verify that the Internet/WAN cloud-facing interfaces are up.
12. Check HQ-ASA outside, inside and DMZ interfaces with show interface ip brief.
13. Confirm IPsec tunnel status for BR1, BR2 and BR3.
14. Confirm OSPF neighbor relationships are FULL.
15. Confirm branch endpoint IP addresses with show ip before testing from HQ.
16. Run one positive test from each branch to HQ.
17. Run one negative test from Guest or ATM to verify policy enforcement.

E.2 Core Commands Used During Validation

Purpose	Command Examples	Interpretation
Interface status	show interface ip brief; show interfaces status	Detect shutdown or disconnected ports
VLAN membership	show vlan brief; show interfaces trunk	Verify access VLANs and trunk allowed lists
Routing	show ip route; show route	Confirm learned branch/HQ routes
OSPF	show ospf neighbor; show ip ospf neighbor	Neighbor FULL means routing exchange is established
IPsec	show crypto ikev1 sa; show crypto ipsec sa	IKE/IPsec SAs and counters validate tunnel traffic
ASA simulation	packet-tracer input ...	Shows ACL, NAT and route decision inside ASA
ARP	show arp; clear arp	Useful for duplicate IP or Layer 2 diagnosis

APPENDIX F - RISK REGISTER

The following risk register summarizes the main technical risks identified in the lab. It can also be used during the presentation to explain why security controls such as segmentation, logging and DMZ isolation were included.

Risk	Affected Area	Impact	Mitigation
Guest device reaches internal banking network	Guest VLAN / branch access	Confidentiality breach and lateral movement	Deny internal 10.0.0.0/8 from Guest VLAN
DMZ web server compromise	DMZ and inside boundary	Attacker pivots from public service to internal servers	Restrict DMZ-to-inside traffic and expose only required ports
ATM network exposure	ATM VLAN	Manipulation of financial transaction devices	Isolate ATM VLAN and permit only required service subnets
Single firewall failure at HQ	HQ perimeter	Loss of branch connectivity and remote access	Future ASA HA pair and redundant WAN
Unmonitored deny events	SOC/logging	Attacks remain invisible	Forward firewall and VPN logs to HQ-SYSLOG-SRV
Addressing conflict	DMZ/endpoints	Unstable ARP and intermittent reachability	IP inventory and ARP verification before testing
Emulator forwarding anomaly	BR2/BR3 routers	False diagnosis of routing/VPN failure	Document no ip cef as lab workaround

APPENDIX G - PRESENTATION DEFENSE NOTES

This appendix prepares short answers for expected jury questions. The answers are intentionally concise and aligned with the final report. They should be used as talking points rather than memorized word-for-word.

G.1 Why is the project not just a simple VPN topology?

The project is not evaluated only by reachability. It combines encrypted branch connectivity, role-based VLAN segmentation, DMZ isolation, ACL policy, negative security tests and centralized logging. Some traffic is expected to work, and some traffic is expected to fail. That distinction is what makes the design closer to a banking security scenario.

G.2 Why are some VLANs shown as design reserve?

The presentation deck contains a scalable production-style VLAN catalog. The final EVE-NG demo implements a subset of that catalog. Reserved VLANs show how the architecture can grow, but the report does not falsely claim that every reserve VLAN has active endpoints in the final topology.

G.3 Why did no ip cef appear in the router configuration?

It was an EVE-NG image-specific workaround. In the lab, route tables, OSPF and IPsec SAs were correct, but transit endpoint traffic did not forward correctly until CEF was disabled. This should not be interpreted as a production recommendation; on real routers, CEF is normally enabled for performance.

G.4 Why is the DMZ important?

The web server is public-facing compared with inside users and protected servers. Placing it in DMZ means that an exposed service does not automatically become part of the trusted inside network. If the web server is attacked, ACLs and firewall zones reduce the possibility of lateral movement.

APPENDIX H - SECURITY POLICY SUMMARY

This appendix summarizes the final security policy in a form suitable for report review. The table focuses on what should be allowed and what should be blocked.

Policy Area	Allowed	Blocked	Reason
Branch-to-HQ	Teller and authorized branch devices to HQ services	Unnecessary guest and unrestricted ATM paths	Support business operations with least privilege
Branch-to-Branch	Authorized operational endpoints through HQ hub	Direct uncontrolled branch mesh	Centralized inspection and routing
Guest Network	Internet path where available	All internal 10.x.x.x networks	Guest devices are untrusted
ATM Network	Required transaction/service servers	General user, guest and arbitrary branch networks	Protect financial transaction devices
DMZ	Inbound web service ports when explicitly configured	DMZ-to-inside unrestricted access	Limit exposure of public services
Management	Admin workstation to device management IPs	Normal users and guest devices to management IPs	Protect control-plane access

APPENDIX I - FINAL CONSISTENCY NOTES

The revised report applies the final topology list as the primary source for site addressing and device inventory. The most important consistency correction is the Satellite Branch 2 address block: it is documented as 10.4.0.0/16. The VPN tunnel transit still uses 10.255.3.0/30, but this is separate from the branch internal address space.

The report also separates the active EVE-NG implementation from the broader future-ready VLAN design. This prevents contradictions between the presentation tables and the visible final topology. In the final report, active devices are listed in the device inventory, while larger VLAN catalogs are described as scalable design capacity where endpoints were not deployed in the live demo.

Finally, the report includes the team members: Kutlu Çağan Akın - 05210000260, Furkan İyem - 05220000316 and Gülnaz Karaca - 05210000266.

APPENDIX J - REVISION LOG

This revision log records the major updates applied to the final report after the last topology photograph, reference archive and presentation PDF were provided. It is included to make the document review process transparent.

Revision Item	Applied Correction
Team members	Added Kutlu Çağan Akin, Furkan İyem and Gülnaz Karaca with student numbers.
Final topology	Aligned the report with the final EVE-NG screenshot and the written topology inventory.
Satellite Branch 2 addressing	Corrected the internal BR3 site block to 10.4.0.0/16.
Presentation alignment	Integrated Zero Trust, Least Privilege, DMZ, IPsec, VLAN segmentation and SOC/syslog themes from the final presentation.
Reference integration	Added the papers and supplemental source list from TEZ_KAYNAK.zip to the reference structure.
VLAN clarity	Separated active implemented VLANs from scalable reserve VLANs to avoid overclaiming.
Configuration appendix	Updated representative configurations with the final site blocks and branch labels.

APPENDIX K – IMAGES AND VERSIONS

Device/Image	Version	Purpose in Project	Topology Devices
Cisco IOU Layer 2 (L2.bin)	IOU L2	Access switching, VLAN configuration, port assignment	ACC-USERS, ACC-PRINTER-CCTV, ACC-DATA-CENTER, BR1-ACC-USERS, BR2-ACC-USERS, BR2-CCTV-ATM, BR3-ACC-USER, ACC-PRINTER-CCTV4
Cisco IOU Layer 3 (L3.bin)	IOU L3	Core switching, inter-VLAN routing	HQ-CORE, BR1-CORE
Cisco ASAv	9.23.1	Firewall, NAT, VPN gateway, Internet edge security	HQ-ASA, BR1-ASA

Ubuntu Server	24.04 LTS	Syslog server and centralized logging services	HQ-SYSLOG-SRV, MAIN_RETAIL_SYSLOG
Ubuntu Server	22.04 LTS	Application and infrastructure servers	WEB-SERVER, APP-SERVER, DB-SERVER
Ubuntu MATE	22.04	End-user workstation simulation and remote worker host	REMOTE-WORKER
Generic Network Devices (EVE-NG built-in nodes)	VPC	ATMs, IP Phones, CCTV Cameras, Printers, Access Point representations	ATMs, VoIP Phones, CCTVs, Printers, Guest-AP, PC-IT-ADMIN, PC-DEVELOPER, PC-FINANCE, PC-HR, PC-MARKETING, PC-DMZ-WEB, PC-TELLERs